Hidden Markov Guided Deep Learning Models for Forecasting Highly Volatile Agricultural Commodity Prices

G. Avinash[a,c], Ramasubramanian V.[b,*], Mrinmoy Ray[c], Ranjit Kumar Paul[c], Samarth Godara[c], Harish Nayak[a,c], Rajeev Ranjan Kumar[c], B. Manjunatha[a,c], Shashi Dahiya[c], Mir Asif Iquebal[c]

[a] *The Graduate School, ICAR - Indian Agricultural Research Institute, New Delhi, India*
[b] *ICAR - National Academy of Agricultural Research Management, Hyderabad, India*
[c] *ICAR - Indian Agricultural Statistics Research Institute, New Delhi, India*

Abstract

Predicting agricultural commodity prices accurately is of utmost importance due to various factors such as perishability, seasonality, production uncertainty *etc.* Moreover, the substantial volatility that may be exhibited in time series further adds to the complexity and constitutes a significant challenge. In this paper, a Hidden Markov (HM) guided Deep Learning (DL) models has been developed on nonlinear and nonstationary price data of agricultural commodities for forecasting by considering some technical indicators *viz.*, Moving Average (MA), Bollinger Bands (BB), Moving Average Convergence Divergence (MACD), Exponential MA (EMA) and Fast Fourier Transformation (FFT). HM Models (HMMs) can effectively handle the sequential dependencies and hidden states, while DL approach can learn complex patterns and relationships within the price series and thus the drawback of lack of generalization capability in the DL model has been overcome by HMM. In this study, the Potato price data of the Champadanga district of West Bengal, India has been utilized to assess the performance of the proposed technique. HMM has been combined with six DL models *viz.*, Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Bidirectional LSTM (BiLSTM) and Bidirectional GRU (BiGRU) for forecast modeling. Performance evaluation metrics *viz.*, Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE) and the insightful Diebold-Mariano (DM) test revealed that Hidden Markov hybridized with DL models surpassed other baseline models in forecasting accuracy for 1-week, 4-week, 8-week and 12-week ahead DL predictions. The proposed approach holds significant promise for enhancing the precision of agricultural commodity price forecasting with far-reaching implications for various stakeholders such as farmers and planners.

Keywords: Bollinger Bands, Recurrent Neural Network (RNN)/ Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM)/ Bidirectional LSTM (BiLSTM), Gated Recurrent Unit (GRU)/ Bidirectional GRU (BiGRU), Fourier Transform

1. Introduction

Precise and timely price forecasting is indispensable for effective planning and strategic decision-making in the case of any commodity or product [1]. Forecasting agricultural commodity prices poses unique challenges compared to non-farm goods and services. These challenges stem from the vulnerability of agricultural commodities to unpredictable events like droughts, floods, disease outbreaks and pest infestations, as well as from factors such as seasonality, demand

*Corresponding author
Email address: r.subramanian@icar.gov.in (Ramasubramanian V.)

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

fluctuations, climate variability, market imperfections, globalization and speculative trading *etc.*[2]. Moreover, the nonlinear and nonstationary characteristics of price data further complicate the process of predicting agricultural commodity prices.

India is the world's second-largest producer of vegetables with a total production of 137.99 million metric tonnes (MMT) compared to China's 600.01 MMT, including significant potato production reaching 56.17 million metric tonnes (MMT) during 2021-22 in India (`www.statista.com`). Vegetable-growing farmers sometimes face challenges even due to overproduction, that may lead to distress sales, crop burning and discarding of produce on roads during bearish market conditions [3]. To address these issues and stabilize prices, it becomes crucial to provide farmers with storage facilities during bearish times, offer guidance on selling strategies during inflationary periods and enhance their knowledge of the supply value chain and provide timely forecasts in one solution. This study utilizes price data from the Champadanga market in West Bengal, India where the thriving potato industry supports the livelihoods of over 60,000 farmers and also contributes area of around 38% of the region's total Potato production.

It has been noticed that, in some cases, models may not perform optimally for price forecasting due to the lack of contextual information, inherent complexities due to the nonlinearity and nonstationarity and limited ability to capture short and long-term fluctuations *etc.* As a solution to this problem, technical indicators provide valuable insights into price trends, volatility and momentum, thereby improving the model's forecasting accuracy for different time horizons making them essential features for more accurate and reliable price predictions [4, 5, 6]. In the present study, technical indicators *viz.,* Moving Average (MA), Bollinger Bands (BB), Moving Average Convergence Divergence (MACD), Exponential Moving Average (EMA) and Fast Fourier Transform (FFT) have been used in the DL models.

Although technical indicators offer valuable insights into price trends and patterns, they may not fully encompass all the intricacies found in price series data. Moreover, DL models are susceptible to overfitting, particularly when working with noisy datasets [7]. Hence DL models can be challenging to train and fine-tune, requiring expertise in hyperparameter optimization and architectural design. To address this, the proposed approach utilizes the strengths of two methodologies *i.e.,* by combining Hidden Markov Model (HMM) with DL and also taking into account technical indicators to analyze underlying patterns in the data series with the aim of overcoming challenges such as overfitting and circumventing local minima traps. In this way, improved data efficiency, better sequential modeling, interpretability and robustness can be achieved.

The paper is structured as follows: section 2 presents the literature review, section 3 discusses related work and background knowledge, section 4 presents the empirical study using a real dataset for the Potato price series and section 5 concludes with findings, remarks and future prospects followed by a list of references.

2. Literature Review

Extensive research has been conducted [8, 9] to enhance price forecasting by employing diverse and advanced time series models [10]. These models can be broadly categorized into two groups: statistical models and artificial intelligence (AI)-based models, both aimed at improving price predictions [11].

The ARIMA model, introduced by Box and Jenkins in 1970, has been widely employed in time series analysis, particularly for forecasting financial data [12, 13, 14, 15, 16]. However, its effectiveness is limited when dealing with nonlinear data. To address this limitation, alternative nonlinear time series models, such as regime-switching models

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

like SETAR [17] and GARCH [18] have emerged. These models can capture nonlinearity but often require specific relationships in the data and may lack generalization ability, as pointed out by [19].

HMMs were developed to address the challenges of complex dependencies and nonlinear relationships in time series data, building on pioneering work by Baum and colleagues [20, 21]. Initially, the HMMs [22] were applied in the speech recognition domain. These models assume that observed data is generated by a Markov process with hidden states, allowing them to capture nonlinearity and temporal dependencies. By modeling hidden states, HMMs extract essential features like trend, seasonality and volatility. However, HMMs' applicability may vary depending on time series characteristics, as chaotic patterns with long-range dependencies may not align with their assumptions [23]. By incorporating market sentiment and volatility patterns, HMMs provide a probabilistic framework to predict price movements, enabling traders, investors, government and farmers to make more informed decisions in uncertain and dynamic market conditions. Additionally, HMMs can handle noisy data, thus enhancing the accuracy and robustness of the forecasting process.

Machine Learning (ML) models have gained prominence for their data-driven nature, interpretability and lack of assumptions [24]. Various ML models, including Artificial Neural Networks (ANNs)/Multilayer Perceptron (MLP) [25], Support Vector Regression (SVR) [26], Random Forest (RF) [27], eXtreme Gradient Boosting (XGBoost) [28], Decision Tree (DT) [29] and ensemble models like stacking [30] and bagging [31] have been utilized for forecasting. ML models offer advantages over traditional model-based approaches due to their adaptability and learning from data. However, ANN may suffer from slow convergence and overfitting [32, 33]. In contrast, DL excels at feature extraction, outperforming other methods when domain knowledge is limited, though it may require a substantial amount of labeled training data for optimal performance.

Deep architectures, achieved by adding extra layers, exploit multiple levels of nonlinear processes, thereby increasing the model's complexity [34]. Deep Neural Networks (DNNs) with more layers efficiently handle complex functions using fewer parameters. These models include Recurrent Neural Networks (RNN), Gated Recurrent Units (GRUs) [35] and Long Short-Term Memory (LSTM) networks [36, 37, 38, 39, 7].

Recent research has witnessed notable studies exploring the integration of HMMs with various DL techniques to enhance time series forecasting accuracy across different domains. For instance, [40] proposed a novel approach combining a Generative Adversarial Network (GAN) with an iteratively refined HMM for unsupervised speech recognition. A hybrid model developed by [41] leveraging HMMs and optimized LSTM networks to predict Bitcoin prices. [42] introduced a neural network-enhanced HMM-based structural time series model tailored for tourism demand forecasting. Building on these advancements, [43] devised an HMM-LSTM model for proactive traffic prediction in 6G wireless networks, while [44] developed a comprehensive GMM-HMM-LSTM model for stock market trend analysis. Additionally, [45] investigated the potential of an HM-BiLSTM-based system for event detection and classification. These studies collectively demonstrate the effectiveness of integrating HMMs with DL techniques to address complex time series forecasting challenges in diverse domains.

This emphasizes the significance of advancing such research progress in the agricultural domain. In this study, we have conducted an investigation on the Potato price series from the Champadanga market, West Bengal, India, employing the technical indicators mentioned earlier, in combination with HMM, to extract relevant features. These extracted features are subsequently fed separately to a range of DL models, *viz.,* RNN, CNN1D, GRU, LSTM, Bidirectional GRU (BiGRU) and Bidirectional LSTM (BiLSTM), with the objective of enhancing forecasting accuracy.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

3. Methodology

3.1. Technical Indicators

The technical indicators *viz.*, Moving Average (MA), Moving Average Convergence Divergence (MACD), Bollinger Bands (BB), Exponential Moving Average (EMA), Momentum and Fast Fourier transformation (FFT) are employed to analyze trends, reduce noise, identify volatility and detect potential price reversal points. Additionally, Fast Fourier Transform (FFT) is used to transform time-domain signals into frequency-domain representations, aiding in the identification of periodic patterns. The integration of these techniques empowers DL models to make more informed predictions, resulting in enhanced accuracy for agricultural price forecasting which can aid farmers and traders in making better decisions.

3.2. Hidden Markov Models (HMMs)

HMMs are probabilistic models that produce a sequence of observations (Y_t) based on a series of underlying hidden states (S_t). They are frequently used to model time-dependent data and have been proven valuable in various domains, including speech recognition, molecular biology and computer vision [46].

HMMs are built upon two fundamental assumptions. Firstly, HMM assumes that an observation at a particular time t, denoted as Y_t, is generated by an underlying process where the corresponding state, S_t, remains hidden from the observer. Secondly, it assumes that this hidden state S_t follows a first-order Markov property, *i.e.,* that the current state S_t, given the previous state S_{t-1}, is independent of all states prior to $t-1$. Likewise, the output of an HMM also adheres to the Markov property. Consequently, the joint distribution of a sequence of hidden states and observations can be factorized as presented by equation (1).

$$P(S_{1:T}, Y_{1:T}) = P(S_1)P(Y_1|S_1)\prod_{t=2}^{T} P(S_t|S_{t-1})P(Y_t|S_t) \tag{1}$$

where $P(S_{1:T}, Y_{1:T})$ represents the joint distribution of the sequence of hidden states ($S_{1:T}$) and observations ($Y_{1:T}$). $P(S_1)$ is the initial probability distribution of the first hidden state S_1. $P(Y_1|S_1)$ is the probability of observing Y_1 given the state S_1. $P(S_t|S_{t-1})$ is the transition probability from state S_{t-1} to state S_t. $P(Y_t|S_t)$ is the probability of observing Y_t given the state S_t. Overall, the equation describes how the joint distribution of hidden states and observations in an HMM can be factorized based on the initial state probability, observation probabilities given the states, and transition probabilities between states. HMM is defined by three key components: A, B, and π, while implicitly determined by the number of observations (N) and the number of hidden states (M) where A represents the state transition probability $M \times M$ matrix, B represents the probability of the observations $M \times N$ matrix, and π is the initial state distribution. Thus, HMM can be defined as: $\lambda = (A, B, \pi)$.

HMMs are utilized to address three fundamental problems, which can be summarized as follows:

1. Problem 1: Given the model $\lambda = (A, B, \pi)$, along with a sequence of observations Y, determine the likelihood of the observed data with respect to the given model through the forward-backward or Expectation-Maximization algorithm.

2. Problem 2: Given the model $\lambda = (A, B, \pi)$, along with a sequence of observations Y, determine the optimal sequence of hidden states underlie the Markov process through the Viterbi algorithm.

3. Problem 3: Given a sequence of observations Y, estimate the parameters of the model, namely A, B and π, through the Baum-Welch algorithm.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

In this study, our approach involves constructing HMM-based models on a given sequence of observations. Subsequently, by calculating the likelihood of the data and determining the optimal sequence of hidden states through the Viterbi algorithm, following the standard methodology, which can be found in, say, [47].

3.3. Deep Learning Models

DL models are at the forefront of AI, mimicking the human brain's neural networks. They excel at representation learning, discovering complex patterns in data without requiring manual feature engineering. The popular and classical models are RNNs, CNN, GRUs, LSTM, BiGRU and BiLSTM models which are considered in our present study and described in subsequently.

3.3.1. Recurrent Neural Networks (RNNs)

RNNs are well-suited for modeling time series data, utilizing interconnected neurons to capture the relationship between past input features and future target variables based on their internal state transitions. However, a significant limitation of RNNs is the gradient vanishing problem, where gradients diminish over time. This limitation can lead to reduced accuracy, especially when dealing with time series data with long-term dependencies. To address this issue, LSTM and GRU networks were introduced as variants of RNNs, offering solutions that allow for better handling of long-term dependencies.

3.3.2. Long Short Term Memory (LSTM)

The limitation of traditional RNNs is addressed [48] in retaining important historical information for long periods by developing the LSTM model. LSTM is an advanced form of RNN specifically designed for sequential data [49]. The mathematical formulation of LSTM is represented in (2,3,4, 5, 6, 7)

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \tag{2}$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{3}$$

$$\tilde{c}_t = \gamma(W_c x_t + U_c h_{t-1} + b_c) \tag{4}$$

$$c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t \tag{5}$$

$$o_t = \sigma(W_0 x_t + U_0 h_{t-1} + b_o) \tag{6}$$

$$h_t = o_t \times \gamma(c_t) \tag{7}$$

The LSTM mechanism is centred around a cell state, denoted as c_t, which serves as a storage unit for information. This information is regulated through three gates: the forget gate (f_t), the input gate (i_t), and the output gate (o_t). The forget gate, as indicated by Eq. (2), decides whether information should be added or omitted. If f_t is close to one (or zero), the information from the input and hidden state will be preserved (or removed) accordingly. The input gate computes an update to the cell state, evaluating the importance of the input for the subsequent cell. Additionally, the output gate generates the output for the hidden states based on Eq. (7). Notably, the activation functions used in LSTM are the hyperbolic tangent function (tanh) and the sigmoid function (σ).

3.3.3. Gated Recurrent Units (GRUs)

GRUs units exhibit similarities to RNN structure. However, the key difference lies in GRUs lacking an output gate and instead employing update and reset gates to manage memory flow, effectively addressing the vanishing

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

gradient problem in RNNs. This gating mechanism allows GRUs to retain long-term information, enabling more accurate predictions. They particularly excel when working with smaller datasets compared to LSTM models [50].

The process can be described in equations (8,9,10, 11) as follows:

$$Z_t = \sigma(x_t w^z + h_{t-1} U^z + b_z) \tag{8}$$

$$r_t = \sigma(x_t w^r + h_{t-1} U^r + b_r) \tag{9}$$

$$\tilde{h}_t = \tanh(r_t \cdot h_{t-1} U + x_t W + b) \tag{10}$$

$$h_t = (1 - Z_t) \cdot \tilde{h}_t + Z_t \cdot h_{t-1} \tag{11}$$

where w^z, w^r, and W denote the weight matrices for the corresponding connected input vectors. U^z, U^r, and U represent the weight matrices of the previous time step and b_z, b_r, and b are biases. The σ denotes the sigmoid function, r_t denotes the reset gate, z_t denotes the update gate and $\tilde{h}_t$ denotes the candidate hidden layer.

3.3.4. Bidirectional LSTM (BiLSTM)

The Bidirectional Long Short-Term Memory (BiLSTM) model has revolutionized neural networks by processing input sequences in both forward and backward directions, preserving information from the future and past simultaneously [51]. It is obtained by combining the forward and the backward hidden states at each time step using a weight matrix and an activation function.

The process can be described in equations (12,13,14) as follows:

$$\overrightarrow{h}_t = \text{LSTM}(x_t, \overrightarrow{h}_{t-1}) \tag{12}$$

$$\overleftarrow{h}_t = \text{LSTM}(x_t, \overleftarrow{h}_{t-1}) \tag{13}$$

$$y_t = \delta(W_{\overrightarrow{h}_y} \overrightarrow{h}_t + W_{\overleftarrow{h}_t} \overleftarrow{h}_t + b_y) \tag{14}$$

3.3.5. Bidirectional GRU (BiGRU)

BiGRU's output layer combines and normalizes results from both layers, enabling the network to extract more information and enhance learning performance. Equations (15, 16, 17, 18, 19) describe the computations within the Bi-GRU network, where $\overrightarrow{h}_t^1$ and $\overrightarrow{h}_t^2$ represent the output vectors of the forward layers at time t in the first and second layers, respectively. Similarly, $\overleftarrow{h}_t^1$ and $\overleftarrow{h}_t^2$ represent the output vectors of the backward layers at the same time t. The mathematical operations and transformations within the BiGRU network facilitate its ability to process and utilize valuable information from both past and future directions, resulting in improved learning capabilities.

$$\overrightarrow{h}_t^1 = f(W_{x\overrightarrow{h}^1} \overrightarrow{x}_t + W_{\overrightarrow{h}^1\overrightarrow{h}^1} \overrightarrow{h}_{t-1}^1 + b_{\overrightarrow{h}^1}) \tag{15}$$

$$\overleftarrow{h}_t^1 = f(W_{x\overleftarrow{h}^1} \overrightarrow{x}_t + W_{\overleftarrow{h}^1\overleftarrow{h}^1} \overleftarrow{h}_{t+1}^1 + b_{\overleftarrow{h}^1}) \tag{16}$$

$$\overrightarrow{h}_t^2 = f(W_{\overrightarrow{h}^1\overrightarrow{h}^2} \overrightarrow{h}_t^1 + W_{\overrightarrow{h}^2\overrightarrow{h}^2} \overrightarrow{h}_{t-1}^2 + b_{\overrightarrow{h}^2}) \tag{17}$$

$$\overleftarrow{h}_t^2 = f(W_{\overleftarrow{h}^1\overleftarrow{h}^2} \overleftarrow{h}_t^1 + W_{\overleftarrow{h}^2\overleftarrow{h}^2} \overleftarrow{h}_{t+1}^2 + b_{\overleftarrow{h}^2}) \tag{18}$$

$$y_t = g(W_{\overrightarrow{h}^2 y} \overrightarrow{h}_t^2 + W_{\overleftarrow{h}^2 y} \overleftarrow{h}_t^2 + b_y) \tag{19}$$

3.3.6. Convolutional Neural Network 1D (CNN1D)

CNN1D utilises a one-dimensional time series as input and uses learnable filters to identify relevant patterns at different time steps. Convolution extracts important features, activation functions add non-linearity, and pooling

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

layers reduce complexity. The processed features are then passed through fully connected layers to refine the information, and the output layer predicts future time steps. This architecture allows the CNN1D to grasp intricate patterns within time series, making it well-suited for forecasting tasks [52].

3.4. Proposed Hidden Markov (HM) based Deep Learning (DL) Modeling

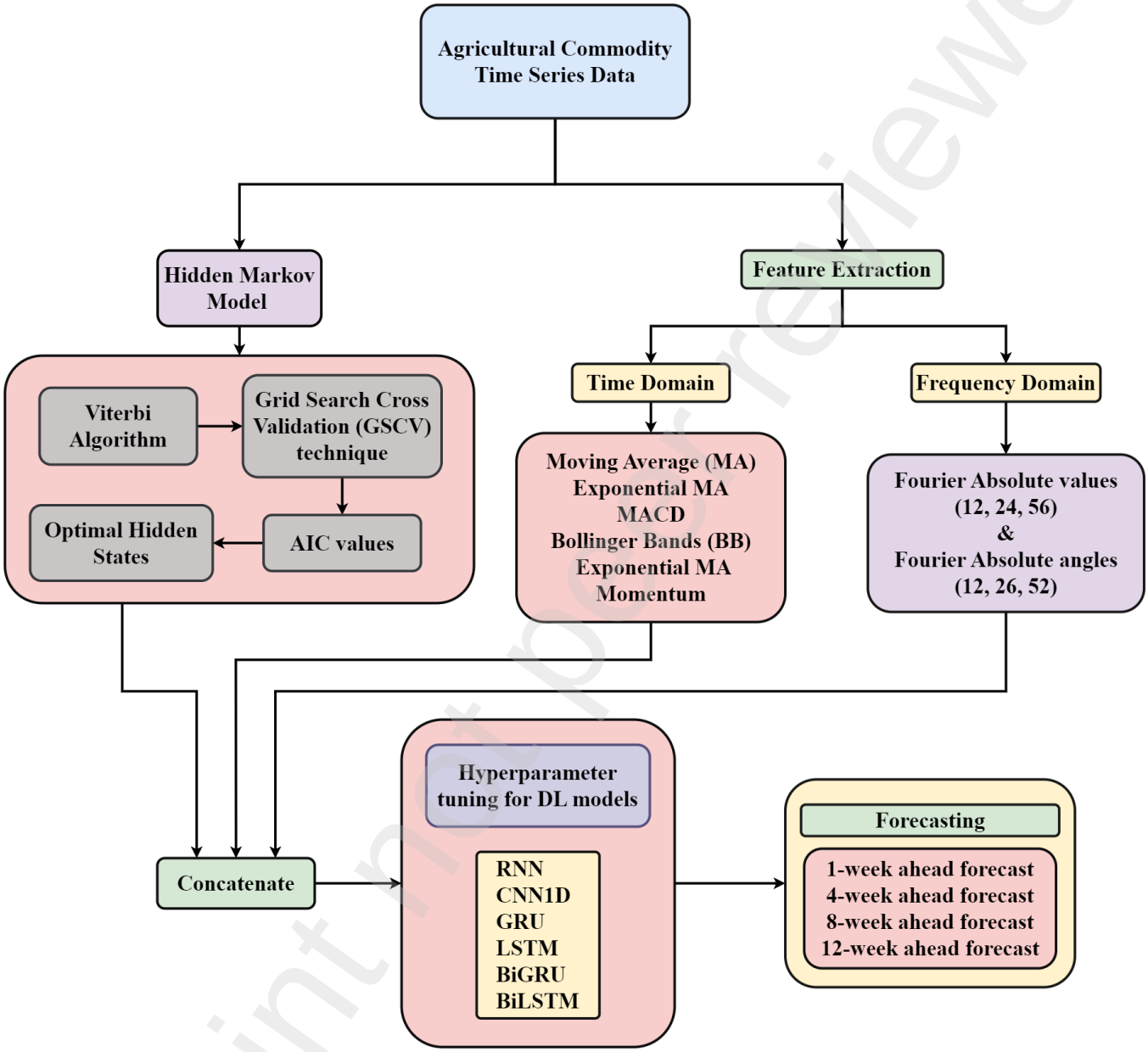


Figure 1: Proposed methodology of HM-DL modeling for price series forecasting

In the proposed approach, we have developed an extensive forecasting framework that leverages technical indicators for extracting features [53]. By integrating Hidden Markov Models (HMMs) into the price series data, our model effectively captures hidden states and dependencies, offering insights into intrinsic market dynamics. Moreover, we have enriched the feature extraction process by incorporating various technical indicators as shown in Figure 1, making the model adaptable to diverse market conditions. We trained multiple models, including conventional DL models *viz.,* RNN, CNN1D, GRU, LSTM, BiGRU, BiLSTM along with HM hybridized models *viz.,* HM-RNN, HM-CNN1D, HM-GRU, HM-LSTM, HM-BiGRU and HM-BiLSTM. To ensure the robustness of our approach, we

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

conducted multi-horizon forecasting (1, 4, 8, 12 weeks) for each model, repeating the process ten times to assess generalization capacity.

To identify the best-performing model, we employed the Diebold-Mariano test for post hoc analysis, allowing us to compare forecast accuracies as depicted in Figure 1. Our methodology not only demonstrates the predictive power of HMMs and technical indicators but also showcases the model's adaptability across different forecasting horizons, with statistically significant results compared to other approaches. The combination of these methodologies results in a sophisticated forecasting tool that has the potential to revolutionize agricultural and financial market analysis, aiding in more informed decision-making.

The analysis was performed using Python software for training DL models. The experiments were executed on a system with an AMD Ryzen 7 5700U processor and 8 GB of RAM, which proved adequate for training and evaluating each DL model. The model processing time varied from 25 to 30 minutes, utilizing the grid search validation (GSCV) technique.

3.5. Forecasting and Evaluation

In order to quantitatively evaluate the performance of each forecasting model for various time horizons, three commonly utilized accuracy metrics were utilized: Root Mean Square Error (RMSE), Mean Absolute Prediction Error (MAPE), and Mean Absolute Error (MAE). Additionally, to identify the superior-performing model, we conducted a post hoc analysis using the Diebold-Mariano test, which allowed us to compare the forecast accuracies effectively.

4. Results and Performance Comparison

4.1. Data Description

This study utilizes the weekly Potato price series (Rs. /Quintal) from January 1, 2006, through May 21, 2023 (obtained from the Agmarknet; `https://agmarknet.gov.in`) from Champadanga market, West Bengal (India). Table 1 displays the descriptive statistics of the Potato series utilized in the study and Figure A.9 exhibits the time plot. The visualizations confirm their nonstationarity and nonlinearity. To further verify this, statistical tests for stationarity and linearity were conducted, as presented in Table 2. The price series for potato ranges from Rs. 194.80 (Rs./Quintal) to Rs. 3578.57 (Rs./Quintal). The standard deviation, reflecting volatility, indicates that series are volatile in nature.

Additionally, the price series is positively skewed and mesokurtic, indicating their non-normality, which is further confirmed by the Shapiro Wilk's test and Jarque–Bera test. Each series consists of 908 observations, and for all models, they are divided into training and testing sets. The training set comprised 90% of the first weekly observations used for model building and in-sample prediction. The remaining 10% of the dataset is used as testing data for model validation and post-sample prediction purposes for 1-week, 4-week, 8-week and 12-week ahead forecasts.

The main objective of this study is to evaluate the performance of the Hidden Markov guided DL models for forecasting agricultural price series, by comparing them with the conventional DL models. The study aims to provide insights into which model is best suited for 1-week, 4-week, 8-week and 12-week forecasting through grid search cross-validation (GSCV) technique.

To evaluate the stationarity and nonlinearity of the series in this study, Augmented Dickey-Fuller (ADF) and Brock-Dechert-Scheinkman (BDS) tests, respectively were performed. The null hypothesis of ADF test states that the series is nonstationary or has a unit root. ADF test confirms the nonstationarity with ADF value 1.42 (p value:

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

Statistic	Value
Minimum (Rs./Quintal)	194.80
Mean (Rs./Quintal)	965.43
Maximum (Rs./Quintal)	3578.57
Standard Deviation (Rs./Quintal)	570.57
Coefficient of Variation (%)	59.13
Skewness	1.25
Kurtosis	1.68
Jarque-Bera test (p - value)	346.59 (<0.001)
Shapiro-Wilk's test (p - value)	0.89 (<0.001)

Table 1: Descriptive statistics of weekly Potato price series

0.21). Whereas, the null hypothesis of the BDS test states that the series follows linearity. The results obtained from the BDS test follow the nonlinearity of the series as shown in Table 2. Moreover, the ARCH LM test reveals long memory and the volatility present in the series.

Epsilon	2 Embedding Dimension	3 Embedding Dimension
0.5 σ	611.93 (< 0.001)	45420.31 (< 0.001)
1 σ	629.31 (< 0.001)	33275.24 (< 0.001)
1.5 σ	482.43 (< 0.001)	13391.53 (< 0.001)
2 σ	496.32 (< 0.001)	16761.26 (< 0.001)

Parenthesis indicates p-values

Table 2: Brock-Dechert-Scheinkman (BDS) test results for nonlinearity

4.2. Technical Indicators

The indicators mentioned in section 3 were utilized as a feature engineering for increasing the forecasting accuracy [54]. Moving Average (MA) was calculated for 7 and 21 weekly data points which help to identify trends and remove noise from data [55]. Then, Exponential Moving Average (EMA) which gives more weight to recent data points, making it more responsive to current trends was also used [56]. Figure 2 depicts the visualization of the last 200 weekly data points visualization for the Bollinger Bands (BB) that act as a volatility indicator by using three bands: an SMA (typically over 20 periods) and two standard deviation bands above and below the moving average. BB helps in understanding the volatility and potential price reversal points in the Potato price series [57]. Moving Average Convergence Divergence (MACD) was also used as a momentum indicator to depict the market's stability through the log momentum [58].

Moreover, the vegetable price series shows underlying periodic patterns or cycles that are not immediately apparent in the time domain [59]. By converting the data to the frequency domain, these periodic components become more visible as peaks at specific frequencies as shown in Figure 3, where Fast Fourier Transformation (FFT) has been used to analyse and decompose the periodicity and seasonality patterns [60, 61]. In our study, 12, 24 and 52 absolute values and angles of Fourier coefficients are captured well in the Potato price periodic patterns.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

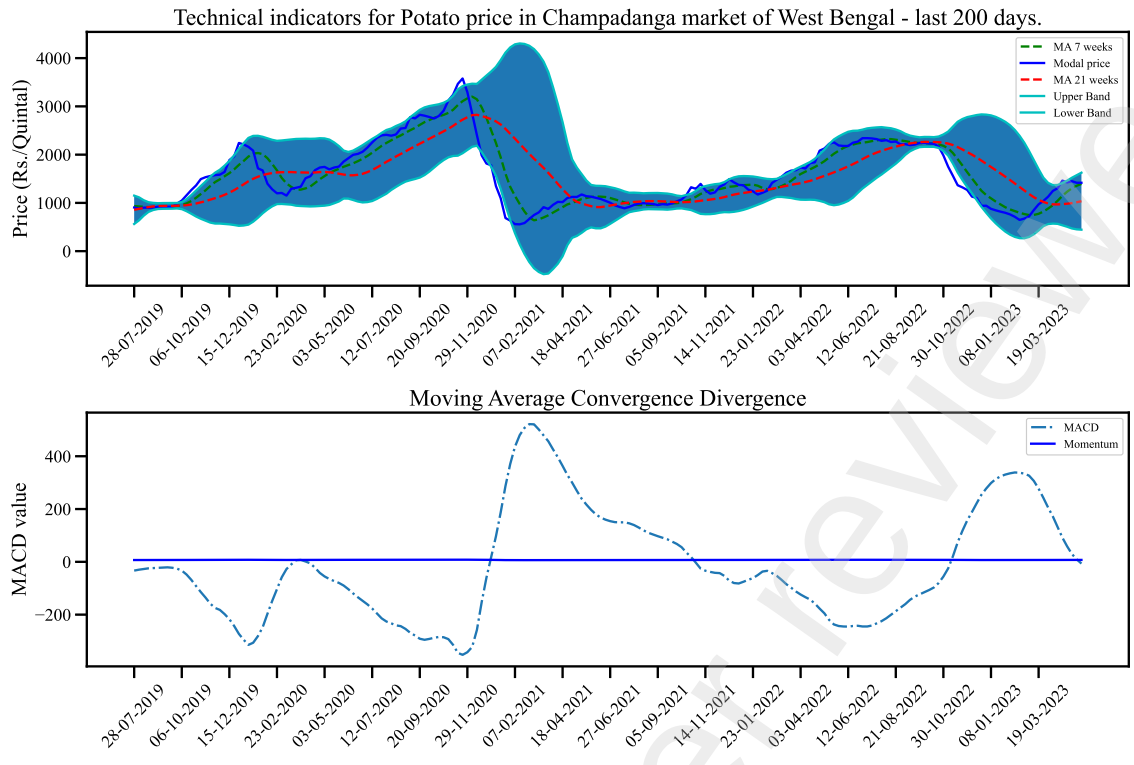


Figure 2: SMA, Bollinger Bands, MACD and Log momentum for the Potato price series

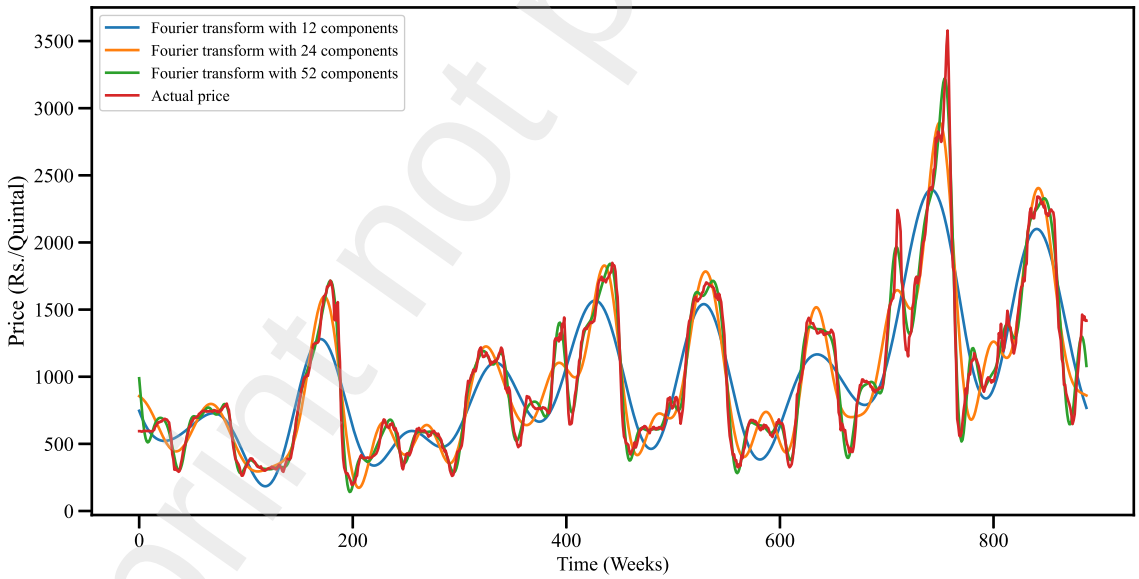


Figure 3: Fourier transformation for extracting 3 months (12 weeks), 6 months (24 weeks) and 12 months (52 weeks) for the Potato price series

Even though technical indicators are good at predicting market volatility, they sometimes fail to adapt to immediate changes in market fluctuations and also have limited predicted power [62]. Hence, Hidden Markov Models (HMMs) were fitted using the Viterbi algorithm (through GSCV) based on AIC values to capture optimal states

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

resulting in capturing the six hidden states for feature engineering as shown in Figure 4. Overall, incorporating HMMs and their associated latent variable representations into DL pipelines provides an effective way of governing agricultural commodity pricing.

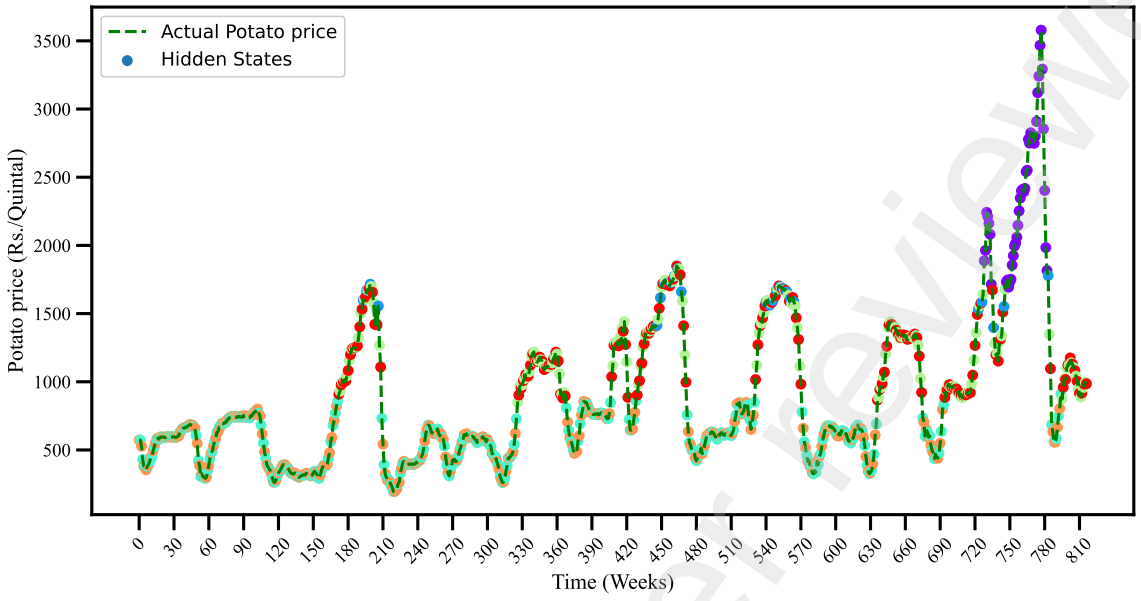


Figure 4: Hidden states were obtained by fitting HMM on Potato price series

Now the final dataset consists of 18 features, *viz.*, Modal price, Minimum and Maximum prices, SMA 7 weeks, SMA 21 weeks, EMA, MACD, Log Momentum, Bollinger bands (20 standard deviations, upper band, lower band), angle of 12, 24 and 52 components, absolute of 12, 24, 52 coefficients from FFT analysis and hidden states obtained from HMM. All features are fed to various DL pipelines to forecast 1-week, 4-week, 8-week and 12-week respectively by following the methodology as shown in the proposed methodology (Figure 1).

4.3. Implementing Forecasting Models

Prior to implementation, the data contained approximately 5-7% missing values. Imputation was performed by taking the average of the preceding and succeeding observations in the weekly data series. To enhance the effectiveness of training DL models, the data were preprocessed using the 'Min-Max Scaler' transformation to normalize it. [53].

After confirming the nonstationarity, nonlinearity, volatility and also extracting features from the technical indicators and hidden states, the above-stated models were ready for training. To enhance the performance of the models considered, hyperparameters were selected among batch sizes (8, 16, 32, 64,128, 256), the number of hidden layers (1, 2, 3), the number of hidden units/ Kernal filters (8, 16, 32, 64, 128, 256, 512). The number of lags was fixed to 12 weeks from the visualisation of the Auto-Correlation Function (ACF) plot and ADF test result. Whereas, the epochs were decided via stopping criteria which are set to 200. These were chosen based on their potential impact on the model's performance and were varied across a range of values in order to find the optimal combination through the Grid search cross-validation technique [7]. Hyperparameters finally selected for the forecasting models considered are given in Supplementary Table A.5.

Later all the above-stated models were trained by using the optimal parameters for 1-week, 4-week, 8-week and 12-week ahead through an early stopping criteria technique to avoid overfitting the model for ten replications to

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

Models	1-Week Ahead Forecasts			4-Week Ahead Forecasts		
	RMSE	**MAE**	**MAPE**	**RMSE**	**MAE**	**MAPE**
RNN	201.80	152.88	9.71	184.06	146.93	10.66
CNN1D	148.74	127.92	8.81	179.82	141.22	10.45
GRU	99.84	76.17	5.13	131.91	106.84	8.12
LSTM	149.51	128.44	9.35	185.28	152.84	10.97
BiGRU	99.84	76.17	5.13	131.91	106.84	8.12
BiLSTM	124.22	105.84	7.78	168.96	140.99	10.69
HM-RNN	137.48	106.47	7.22	182.26	145.89	10.31
HM-CNN1D	159.34	114.53	7.98	164.82	132.45	9.86
HM-GRU	79.69	61.72	4.30	113.86	91.29	6.49
HM-LSTM	164.45	124.13	9.03	150.59	127.46	9.88
HM-BiGRU	91.66	72.20	5.33	125.09	100.82	7.24
HM-BiLSTM	122.31	103.77	7.52	154.18	130.14	9.93

Table 3: Models performance for 1-week and 4-week ahead forecast

Models	8-Week Ahead Forecasts			12-Week Ahead Forecasts		
	RMSE	**MAE**	**MAPE**	**RMSE**	**MAE**	**MAPE**
RNN	200.22	158.78	11.60	233.61	177.53	13.48
CNN1D	194.74	148.95	11.30	219.76	169.47	12.59
GRU	162.96	126.85	9.23	261.82	196.28	14.20
LSTM	202.50	165.52	12.56	276.61	221.71	17.51
BiGRU	202.50	165.52	12.56	276.61	221.71	17.51
BiLSTM	207.31	163.22	11.77	239.04	183.68	13.43
HM-RNN	194.87	151.86	11.05	214.32	176.26	12.61
HM-CNN1D	179.54	140.07	10.29	213.77	161.84	11.94
HM-GRU	157.07	123.60	9.10	232.34	172.77	12.63
HM-LSTM	195.18	154.50	12.18	251.35	196.24	15.28
HM-BiGRU	191.74	153.48	11.09	182.19	151.52	11.18
HM-BiLSTM	185.70	146.36	10.61	235.15	184.23	12.71

Table 4: Models performance for 8-week and 12-week ahead forecast

stabilize the model and forecast on the test data. Table 3 presents the evaluation metrics (RMSE, MAE and MAPE) for various models used in both one-week ahead and four-week ahead forecasts. HM-GRU model performed best with the lowest RMSE (79.69), MAE (61.72), and MAPE (4.30), which outperformed traditional DL models for a week ahead forecast. Similarly, for medium-term i.e., four weeks ahead forecast HM-GRU model has the lowest RMSE (113.86), MAE (91.29) and MAPE (6.49) values. The bidirectional variants (BiGRU and BiLSTM) exhibit similar performance to their regular counterparts (GRU and LSTM) for both forecasting horizons. These results highlight the effectiveness of Hidden Markov-based GRU for short and medium-term forecasts, respectively.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

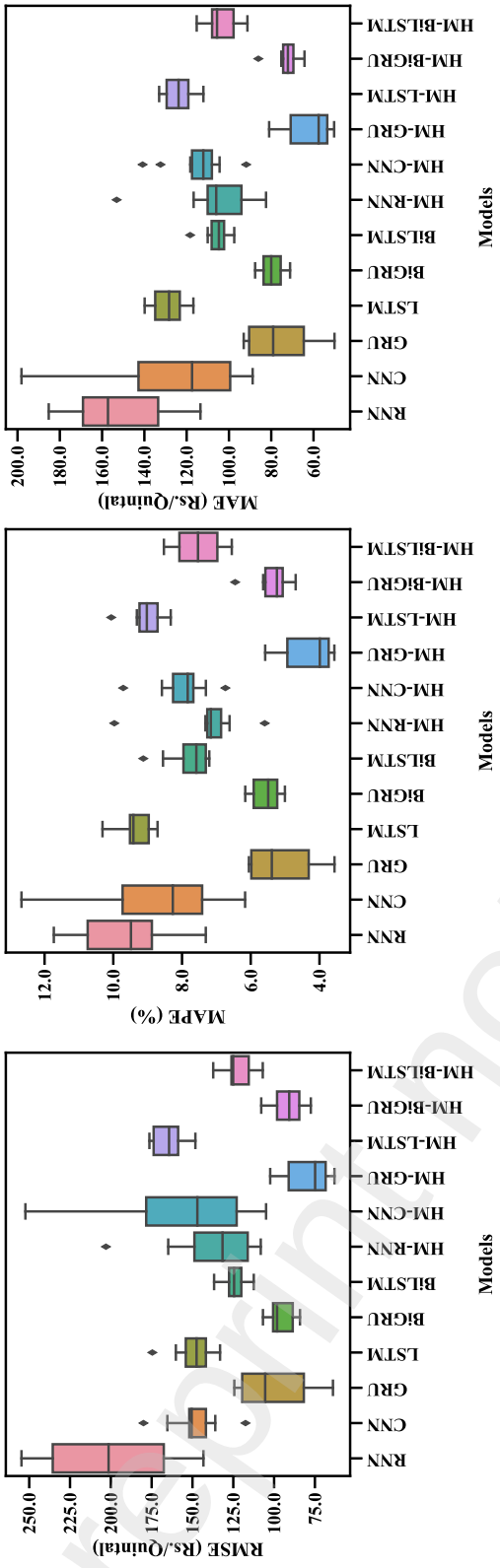


Figure 5: Visualization of model's stability on test data for 1-week ahead forecast

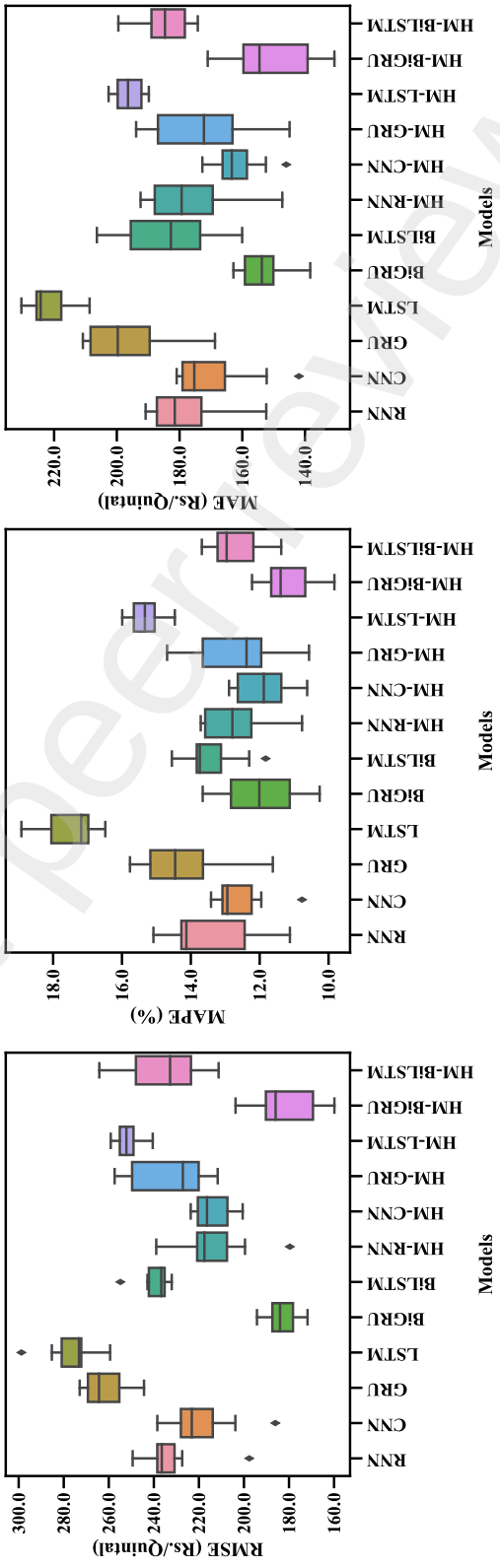


Figure 6: Visualization of model's stability on test data for 12-week ahead forecast

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

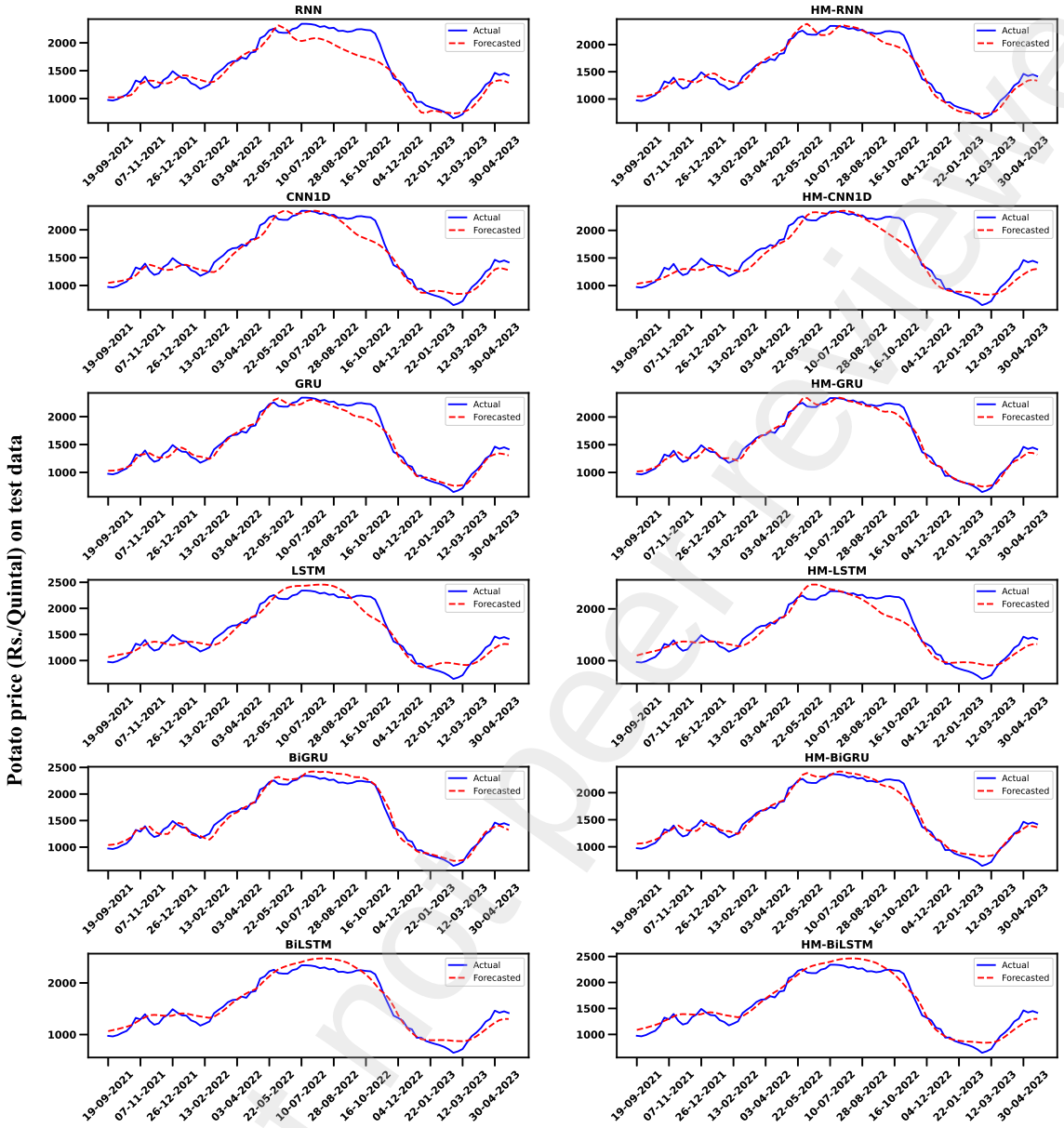


Figure 7: 1-week ahead forecasted price series obtained from DL and HM-DL models

For the 8-week ahead forecasts, the HM-GRU model had the lowest RMSE (157.07), MAE (123.60) and MAPE (9.10) which outperformed other models as given in Table 4. For the 12-week ahead forecasts, the HM-BiGRU model performed exceptionally well with the lowest RMSE (182.19), MAE (151.52) and MAPE (11.18). The results also highlight the effectiveness of HM-GRU and HM-BiGRU models for medium and long-term forecasts, respectively. However, the HM-BiGRU model stands out for the twelve-week ahead forecast, showcasing its capability for more extended prediction horizons. Moreover, the choice of the most suitable model should be made based on the specific forecasting requirements and the length of the prediction horizon.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

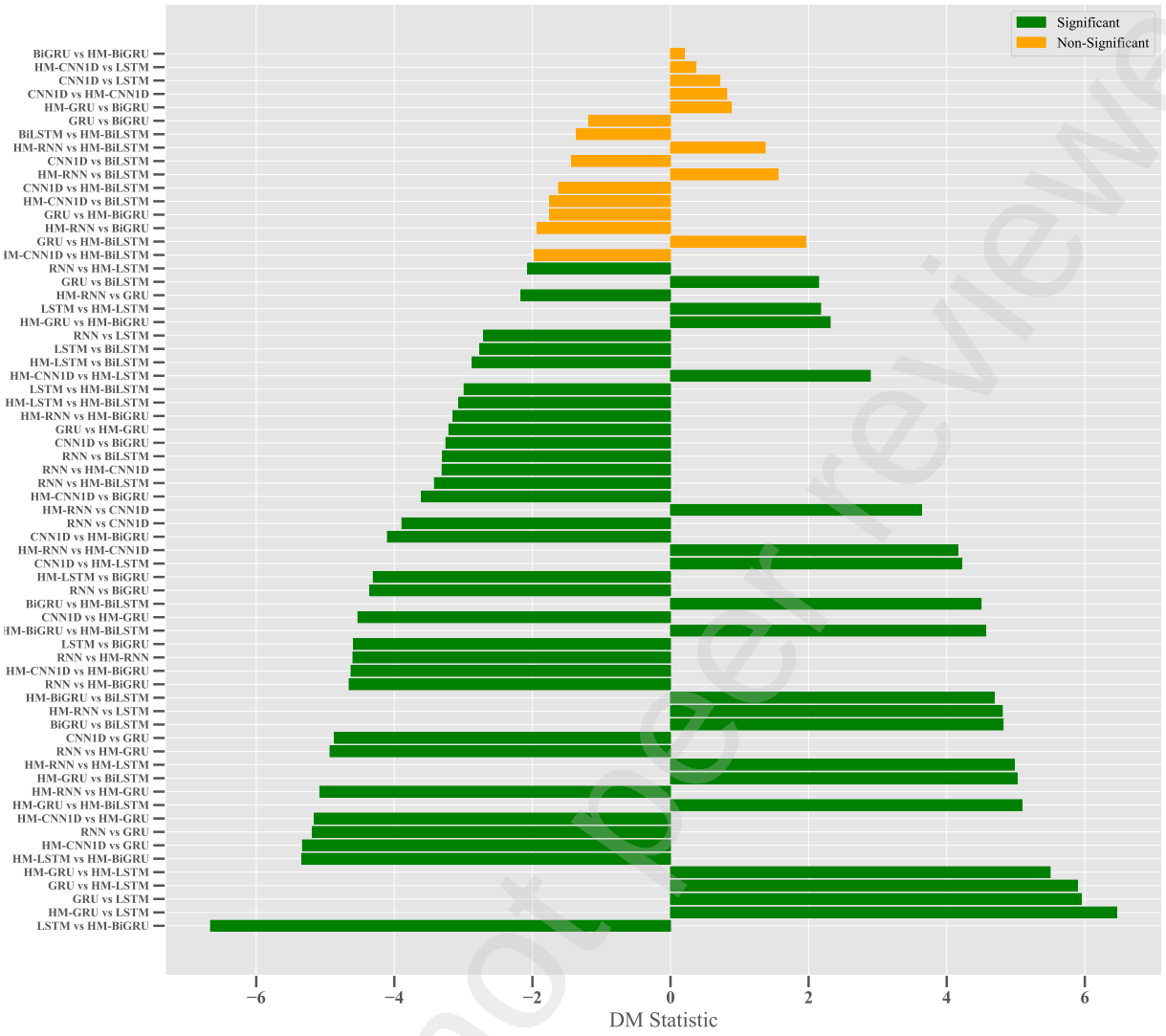


Figure 8: DM test on Potato price series for 1-week ahead forecasts

5. Discussion

The evaluation of HM-based DL models, namely HM-RNN, HM-CNN1D, HM-GRU, HM-LSTM, HM-BiGRU, and HM-BiLSTM, against their DL counterparts (RNN, CNN1D, GRU, LSTM, BiGRU, and BiLSTM) consistently shows that at least 3 (sometimes 4) HM-based models perform better. Figure 7 demonstrates that the HM-based DL models produce forecasted values closer to the actual data series for most time periods in the test data compared to the baseline DL model. Similar performance is observed for 4-week, 8-week, and 12-week ahead forecasts in Figures A.12, A.14 and A.16.

Boxplots in Figure 5, A.10, A.11 and 6 illustrate evaluation metrics for various forecast horizons. The interquartile range widens as the forecast horizon increases. Notably, HM-based models exhibit smaller boxplots for 1 and 4-week

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

ahead forecasts (as shown in Figure 5 and A.10), indicating greater consistency compared to the baseline DL models. A similar trend is observed for 8-weeks ahead forecasts (Figure A.11), except for the HM-LSTM model. However, no discernible patterns could be observed for the 12-week ahead forecasts (Figure 6). This analysis suggests that the HM-based model demonstrates improved performance and consistency over the baseline DL models, especially for short to medium-term forecast horizons. Moreover, the relationship between the evaluation metrics and the multi-horizon ahead forecast is visualized in Figures A.18, A.19 and A.20.

Upon careful examination of the figures, it becomes evident that the forecasts generated by the HM-based DL models, such as HM-RNN, HM-LSTM, and HM-BiGRU, differ significantly from their corresponding standalone DL models, *i.e.*, RNN, LSTM and BiGRU. Notably, HM-BiGRU stood out as the superior model for most forecasting scenarios, except for 8-weeks ahead forecasts, as depicted in Figure 8, A.13, A.15 and A.17. However, the comparison also revealed that the forecasts of HM-CNN1D did not exhibit significant differences across all multi-horizon situations. Specifically, 50, 30, 22, and 35 pairs were found to be significantly different for 1-week, 4-week, 8-week, and 12-week forecasts, respectively. On average, nearly half of the pairwise comparisons showed significant distinctions. This study highlights the importance of HMM in capturing hidden states, handling noisy data, modeling short dependencies, and identifying chaotic patterns, which in turn benefits the DL model's performance. Consequently, the fusion of HMM and DL leverages the strengths of each approach, leading to improved overall performance compared to using DL models alone.

6. Conclusion

This study introduces an innovative approach that enhances agricultural commodity price forecasting precision by combining Deep Learning (DL), hidden states obtained by HMM and several technical indicators. The HMM's hidden states are utilized as a powerful feature extraction technique and integrated into six baseline DL models. The results demonstrate a significant improvement in forecasting accuracy compared to traditional DL models, as validated by evaluation metrics and the Diebold-Mariano (DM) posthoc test on price series data. Remarkably, the proposed HM-DL models achieve these superior results without incurring additional computational time during training. The findings underscore the effectiveness of this hybrid approach over existing methods. The proposed methodology demonstrates the applicability across diverse domains, encompassing volatile data present in agricultural or financial markets. Future research avenues can explore the inclusion of weather conditions, market demand, economic indicators and sentiment analysis to extract the news from social media platforms for the enhancement of forecasting accuracy. Furthermore, the application of HMM-DL applied in diverse sectors like finance, energy and healthcare promises potential benefits in making well-informed decisions based on complex time series data.

Acknowledgments

The facilities provided by The Graduate School, ICAR-Indian Agricultural Research Institute (ICAR-IARI) and ICAR-Indian Agricultural Statistics Research Institute (ICAR-IASRI) in the form of institute fellowship as an ICAR-SRF fellowship under grant number (USID: U-18-KA-01-002-M-A-87) is duly acknowledged for carrying out this study, which is part of his Doctoral research work carried out at ICAR-IASRI, New Delhi, India.

Supplementary Information

Supplementary sheet attached

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

References

[1] T. Xiong, C. Li, Y. Bao, Seasonal forecasting of agricultural commodity price using a hybrid STL and ELM method: Evidence from the vegetable market in China, Neurocomputing 275 (2018) 2831–2844. doi:10.1016/j.neucom.2017.11.053.

[2] H. Yin, D. Jin, Y. H. Gu, C. J. Park, S. K. Han, S. J. Yoo, STL-ATTLSTM: Vegetable Price Forecasting Using STL and Attention Mechanism-Based LSTM, Agriculture 10 (12) (2020) 612. doi:10.3390/agriculture10120612.

[3] in: A. Gulati, K. Ganguly, H. Wardhan (Eds.), Tomato, Onion and Potato (TOP) Value Chains.In Agricultural Value Chains in India: Ensuring Competitiveness, Inclusiveness, Sustainability, Scalability, and Improved Finance, India Studies in Business and Economics, Springer Nature Singapore, Singapore, 2022, pp. 33–97. doi:10.1007/978-981-33-4268-2.

[4] J. Wang, J. Kim, Predicting Stock Price Trend Using MACD Optimized by Historical Volatility, Mathematical Problems in Engineering 2018 (2018) 1–12. doi:10.1155/2018/9280590.

[5] Y. Shynkevich, T. McGinnity, S. A. Coleman, A. Belatreche, Y. Li, Forecasting price movements using technical indicators: Investigating the impact of varying input window length, Neurocomputing 264 (2017) 71–88. doi:10.1016/j.neucom.2016.11.095.

[6] W. Yang, J. Wang, T. Niu, P. Du, A novel system for multi-step electricity price forecasting for electricity market management, Applied Soft Computing 88 (2020) 106029. doi:10.1016/j.asoc.2019.106029.

[7] K. N. Singh, K. Sharma, G. Avinash, R. R. Kumar, M. Ray, V. Ramasubramanian, A. Lama, S. B. Lal, LSTM based Stacked Autoencoder Approach for Time Series Forecasting, Journal of the Indian Society of Agricultural Statistics 77 (2023) 71–78.
URL http://isas.org.in/isa/volume/10-KNSingh.pdf

[8] O. B. Sezer, M. U. Gudelek, A. M. Ozbayoglu, Financial time series forecasting with deep learning : A systematic literature review: 2005–2019, Applied Soft Computing 90 (2020) 106181. doi:10.1016/j.asoc.2020.106181.

[9] A. M. Ozbayoglu, M. U. Gudelek, O. B. Sezer, Deep learning for financial applications : A survey, Applied Soft Computing 93 (2020) 106384. doi:10.1016/j.asoc.2020.106384.

[10] L. Wang, J. Feng, X. Sui, X. Chu, W. Mu, Agricultural product price forecasting methods: Research advances and trend, British Food Journal 122 (7) (2020) 2121–2138. doi:10.1108/BFJ-09-2019-0683.

[11] L. Yu, Y. Zhao, L. Tang, Ensemble Forecasting for Complex Time Series Using Sparse Representation and Neural Networks: Sparse representation for complex time series forecasting, Journal of Forecasting 36 (2) (2017) 122–138. doi:10.1002/for.2418.

[12] C. Kocak, ARMA(p,q) type high order fuzzy time series forecast method based on fuzzy logic relations, Applied Soft Computing 58 (2017) 92–103. doi:10.1016/j.asoc.2017.04.021.

[13] A. A. Adebiyi, A. O. Adewumi, C. K. Ayo, Comparison of ARIMA and Artificial Neural Networks Models for Stock Price Prediction, Journal of Applied Mathematics 2014 (2014) 1–7. doi:10.1155/2014/614342.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

[14] A. A. Ariyo, A. O. Adewumi, C. K. Ayo, Stock Price Prediction Using the ARIMA Model, in: 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation, IEEE, Cambridge, United Kingdom, 2014, pp. 106–112. doi:10.1109/UKSim.2014.67.

[15] J. E. Jarrett, E. Kyper, ARIMA Modeling with Intervention to Forecast and Analyze Chinese Stock Prices, International Journal of Engineering Business Management 3 (2011) 17. doi:10.5772/50938.

[16] Avinash,G., Ramasubramanian, V., Gopalakrishnan, B. N., Heterogeneous Auto-Regressive Modeling based Realised Volatility Forecasting., Statistics and Applications, to appear. (2022).

[17] S. Mehdizadeh, F. Fathian, J. F. Adamowski, Hybrid artificial intelligence-time series models for monthly streamflow modeling, Applied Soft Computing 80 (2019) 873–887. doi:10.1016/j.asoc.2019.03.046.

[18] Z. Lin, Modelling and forecasting the stock market volatility of SSE Composite Index using GARCH models, Future Generation Computer Systems 79 (2018) 960–972. doi:10.1016/j.future.2017.08.033.

[19] R. Weron, Electricity price forecasting: A review of the state-of-the-art with a look into the future, International Journal of Forecasting 30 (4) (2014) 1030–1081. doi:10.1016/j.ijforecast.2014.08.008.

[20] L. E. Baum, T. Petrie, Statistical Inference for Probabilistic Functions of Finite State Markov Chains, The Annals of Mathematical Statistics 37 (6) (1966) 1554–1563. doi:10.1214/aoms/1177699147.

[21] L. E. Baum, G. Sell, Growth transformations for functions on manifolds, Pacific Journal of Mathematics 27 (2) (1968) 211–227. doi:10.2140/pjm.1968.27.211.

[22] L. Rabiner, A tutorial on hidden Markov models and selected applications in speech recognition, Proceedings of the IEEE 77 (2) (Feb./1989) 257–286. doi:10.1109/5.18626.

[23] M. Awad, R. Khanna, Efficient Learning Machines: Theories, Concepts, and Applications for Engineers and System Designers, Apress, Berkeley, CA, 2015. doi:10.1007/978-1-4302-5990-9.

[24] S. Makridakis, E. Spiliotis, V. Assimakopoulos, Statistical and Machine Learning forecasting methods: Concerns and ways forward, PLOS ONE 13 (3) (2018) e0194889. doi:10.1371/journal.pone.0194889.

[25] s. Haykin, Neural Networks and Learning Machines, 3rd Edition, Pearson Education India, 2009.

[26] B. M. Henrique, V. A. Sobreiro, H. Kimura, Stock price prediction using support vector regression on daily and up to the minute prices, The Journal of Finance and Data Science 4 (3) (2018) 183–201. doi:10.1016/j.jfds.2018.04.003.

[27] I. K. Nti, A. F. Adekoya, B. A. Weyori, Random Forest Based Feature Selection of Macroeconomic Variables for Stock Market Prediction, American Journal of Applied Sciences 16 (7) (2019) 200–212. doi:10.3844/ajassp.2019.200.212.

[28] S. Basak, S. Kar, S. Saha, L. Khaidem, S. R. Dey, Predicting the direction of stock market prices using tree-based classifiers, The North American Journal of Economics and Finance 47 (2019) 552–567. doi:10.1016/j.najef.2018.06.013.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

[29] Binoy.B. Nair, V. Mohandas, N. R. Sakthivel, A Decision tree- Rough set Hybrid System for Stock Market Trend Prediction, International Journal of Computer Applications 6 (9) (2010) 1–6. doi:10.5120/1106-1449.

[30] M. Jiang, J. Liu, L. Zhang, C. Liu, An improved Stacking framework for stock index prediction by leveraging tree-based ensemble models and deep learning algorithms, Physica A: Statistical Mechanics and its Applications 541 (2020) 122272. doi:10.1016/j.physa.2019.122272.

[31] H. Wang, Y. Jiang, H. Wang, Stock return prediction based on Bagging-decision tree, in: 2009 IEEE International Conference on Grey Systems and Intelligent Services (GSIS 2009), IEEE, Nanjing, China, 2009, pp. 1575–1580. doi:10.1109/GSIS.2009.5408165.

[32] E. Guresen, G. Kayakutlu, T. U. Daim, Using artificial neural network models in stock market index prediction, Expert Systems with Applications 38 (8) (2011) 10389–10397. doi:10.1016/j.eswa.2011.02.068.

[33] C. Wang, H. Yang, C. Bartz, C. Meinel, Image Captioning with Deep Bidirectional LSTMs, in: Proceedings of the 24th ACM International Conference on Multimedia, ACM, Amsterdam The Netherlands, 2016, pp. 988–997. doi:10.1145/2964284.2964299.

[34] J. F. Torres, D. Hadjout, A. Sebaa, F. Martínez-Álvarez, A. Troncoso, Deep Learning for Time Series Forecasting: A Survey, Big Data 9 (1) (2021) 3–21. doi:10.1089/big.2020.0159.

[35] K. A. Althelaya, E.-S. M. El-Alfy, S. Mohammed, Stock Market Forecast Using Multivariate Analysis with Bidirectional and Stacked (LSTM, GRU), in: 2018 21st Saudi Computer Society National Computer Conference (NCC), IEEE, Riyadh, 2018, pp. 1–7. doi:10.1109/NCG.2018.8593076.

[36] R. Jaiswal, G. K. Jha, R. R. Kumar, K. Choudhary, Deep long short-term memory based model for agricultural price forecasting, Neural Computing and Applications 34 (6) (2022) 4661–4676. doi:10.1007/s00521-021-06621-3.

[37] S. Zaheer, N. Anjum, S. Hussain, A. D. Algarni, J. Iqbal, S. Bourouis, S. S. Ullah, A Multi Parameter Forecasting for Stock Time Series Data Using LSTM and Deep Learning Model, Mathematics 11 (3) (2023) 590. doi:10.3390/math11030590.

[38] M. Heidarpanah, F. Hooshyaripor, M. Fazeli, Daily electricity price forecasting using artificial intelligence models in the Iranian electricity market, Energy 263 (2023) 126011. doi:10.1016/j.energy.2022.126011.

[39] N. Latif, J. D. Selvam, M. Kapse, V. Sharma, V. Mahajan, Comparative Performance of LSTM and ARIMA for the Short-Term Prediction of Bitcoin Prices, Australasian Accounting, Business and Finance Journal 17 (1) (2023) 256–276. doi:10.14453/aabfj.v17i1.15.

[40] K.-Y. Chen, C.-P. Tsai, D.-R. Liu, H.-Y. Lee, L.-s. Lee, Completely Unsupervised Speech Recognition By A Generative Adversarial Network Harmonized With Iteratively Refined Hidden Markov Models (2019). doi:10.48550/ARXIV.1904.04100.

[41] I. A. Hashish, F. Forni, G. Andreotti, T. Facchinetti, S. Darjani, A Hybrid Model for Bitcoin Prices Prediction using Hidden Markov Models and Optimized LSTM Networks, in: 2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA), IEEE, Zaragoza, Spain, 2019, pp. 721–728. doi:10.1109/ETFA.2019.8869094.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

[42] Y. Yao, Y. Cao, A Neural network enhanced hidden Markov model for tourism demand forecasting, Applied Soft Computing 94 (2020) 106465. doi:10.1016/j.asoc.2020.106465.

[43] Y. Peng, T. Feng, C. Yang, C. Leng, L. Jiao, X. Zhu, L. Cao, R. Li, HMM-LSTM for Proactive Traffic Prediction in 6G Wireless Networks, in: 2021 IEEE 21st International Conference on Communication Technology (ICCT), IEEE, Tianjin, China, 2021, pp. 544–548. doi:10.1109/ICCT52962.2021.9657967.

[44] M. Liu, J. Huo, Y. Wu, J. Wu, Stock Market Trend Analysis Using Hidden Markov Model and Long Short Term Memory (Apr. 2021). arXiv:2104.09700.

[45] M. I. Khan, B. Acharya, R. K. Chaurasiya, Hybrid BiLSTM-HMM based event detection and classification system for food intake recognition, in: 2022 First International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT), IEEE, Trichy, India, 2022, pp. 1–5. doi:10.1109/ICEEICT53079.2022.9768487.

[46] Z. Ghahramani, An Introduction to Hidden Markov Models and Bayesian Networks, International Journal of Pattern Recognition and Artificial Intelligence 15 (01) (2001) 9–42. doi:10.1142/S0218001401000836.

[47] P. Giudici, I. Abu Hashish, A hidden Markov model to detect regime changes in cryptoasset markets, Quality and Reliability Engineering International 36 (6) (2020) 2057–2065. doi:10.1002/qre.2673.

[48] S. Hochreiter, J. Schmidhuber, Long Short-Term Memory, Neural Computation 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[49] M. Z. Alom, T. M. Taha, C. Yakopcic, S. Westberg, P. Sidike, M. S. Nasrin, M. Hasan, B. C. Van Essen, A. A. S. Awwal, V. K. Asari, A State-of-the-Art Survey on Deep Learning Theory and Architectures, Electronics 8 (3) (2019) 292. doi:10.3390/electronics8030292.

[50] J. Chung, C. Gulcehre, K. Cho, Y. Bengio, Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling (Dec. 2014). arXiv:1412.3555.

[51] Y. Huang, X. Dai, Q. Wang, D. Zhou, A hybrid model for carbon price forecasting using GARCH and long short-term memory network, Applied Energy 285 (2021) 116485. doi:10.1016/j.apenergy.2021.116485.

[52] Y.-C. Chen, W.-C. Huang, Constructing a stock-price forecast CNN model with gold and crude oil indicators, Applied Soft Computing 112 (2021) 107760. doi:10.1016/j.asoc.2021.107760.

[53] M. Ahsan, M. Mahmud, P. Saha, K. Gupta, Z. Siddique, Effect of Data Scaling Methods on Machine Learning Algorithms and Model Performance, Technologies 9 (3) (2021) 52. doi:10.3390/technologies9030052.

[54] M. Tanaka-Yamawaki, S. Tokuoka, Adaptive use of technical indicators for the prediction of intra-day stock prices, Physica A: Statistical Mechanics and its Applications 383 (1) (2007) 125–133. doi:10.1016/j.physa.2007.04.126.

[55] J. M.-J. Leung, T. T.-L. Chong, An empirical comparison of moving average envelopes and Bollinger Bands, Applied Economics Letters 10 (6) (2003) 339–341. doi:10.1080/1350485022000041032.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

[56] A. Raudys, V. Lenčiauskas, E. Malčius, Moving Averages for Financial Data Smoothing, in: T. Skersys, R. Butleris, R. Butkiene (Eds.), Information and Software Technologies, Vol. 403, Springer Berlin Heidelberg, Berlin, Heidelberg, 2013, pp. 34–45. doi:10.1007/978-3-642-41947-8_4.

[57] M.-Y. Day, Y. Cheng, P. Huang, Y. Ni, The profitability of Bollinger Bands trading bitcoin futures, Applied Economics Letters 30 (11) (2023) 1437–1443. doi:10.1080/13504851.2022.2060494.

[58] R. Vaidya, Moving Average Convergence-Divergence (MACD) Trading Rule: An Application in Nepalese Stock Market "NEPSE", Quantitative Economics and Management Studies 1 (6) (2020) 366–374. doi:10.35877/454RI.qems197.

[59] C. S. C. Sekhar, D. Roy, Y. Bhatt, Food inflation and volatility in India: Trends and determinants, Indian Economic Review 53 (1-2) (2018) 65–91. doi:10.1007/s41775-018-0017-z.

[60] K. Yi, Q. Zhang, S. Wang, H. He, G. Long, Z. Niu, Neural Time Series Analysis with Fourier Transform: A Survey (2023). doi:10.48550/ARXIV.2302.02173.

[61] R. Li, P. Jiang, H. Yang, C. Li, A novel hybrid forecasting scheme for electricity demand time series, Sustainable Cities and Society 55 (2020) 102036. doi:10.1016/j.scs.2020.102036.

[62] M. Göçken, M. Özçalıcı, A. Boru, A. T. Dosdoğru, Integrating metaheuristics and Artificial Neural Networks for improved stock price prediction, Expert Systems with Applications 44 (2016) 320–331. doi:10.1016/j.eswa.2015.09.029.

Appendix A. Supplimentary figures

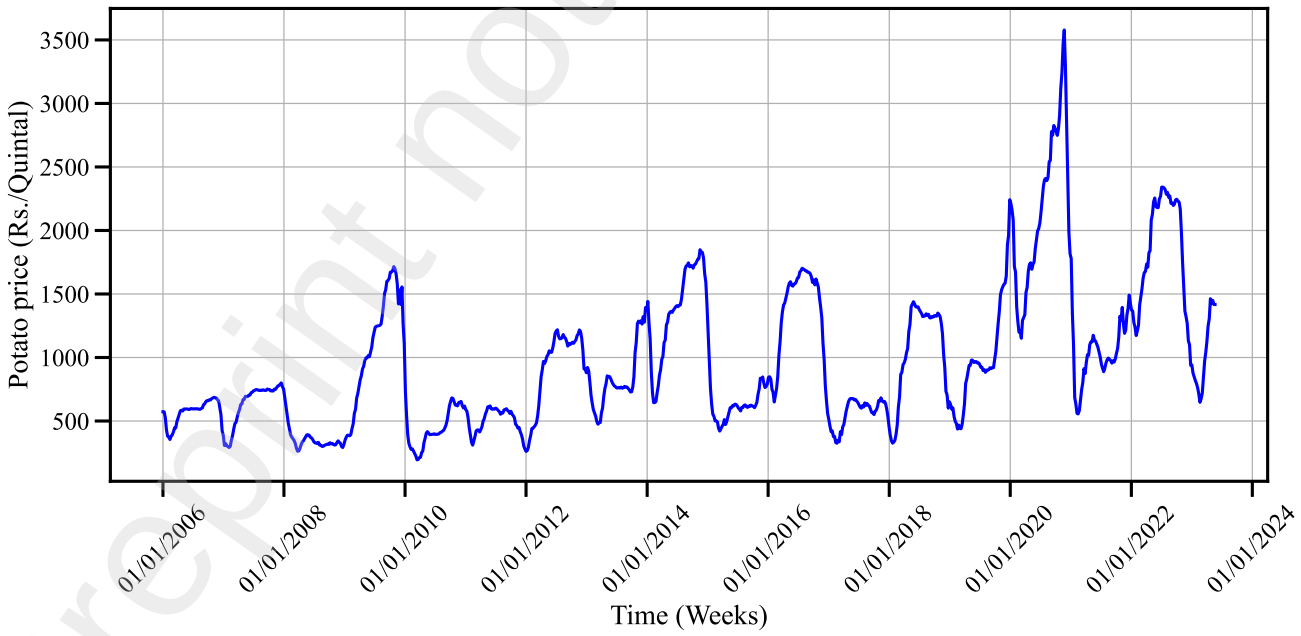


Figure A.9: Time series plot of weekly potato price series

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

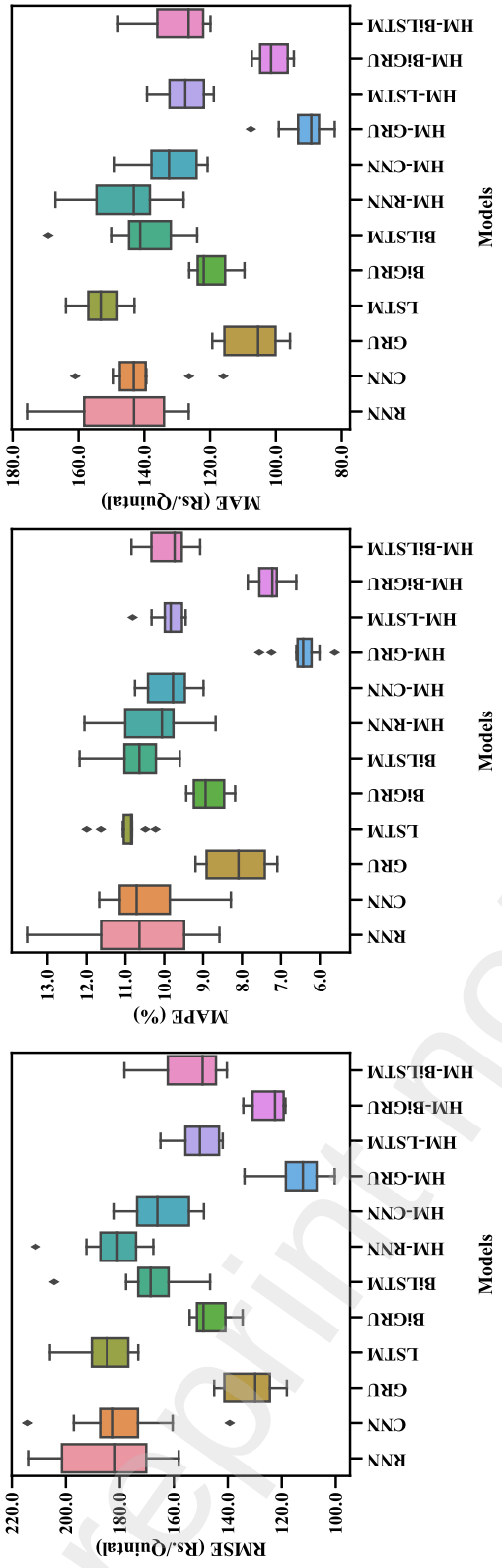


Figure A.10: Visualization of various model's stability on test data for 4 weeks ahead forecasts

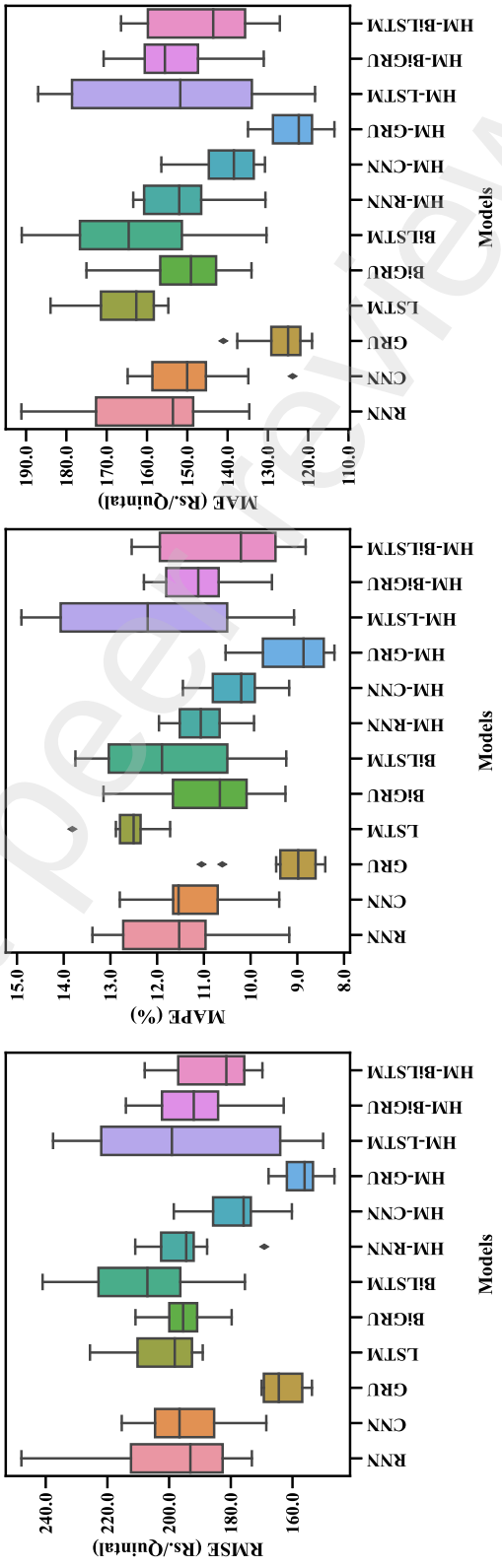


Figure A.11: Visualization of various model's stability on test data for 8 weeks ahead forecasts

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

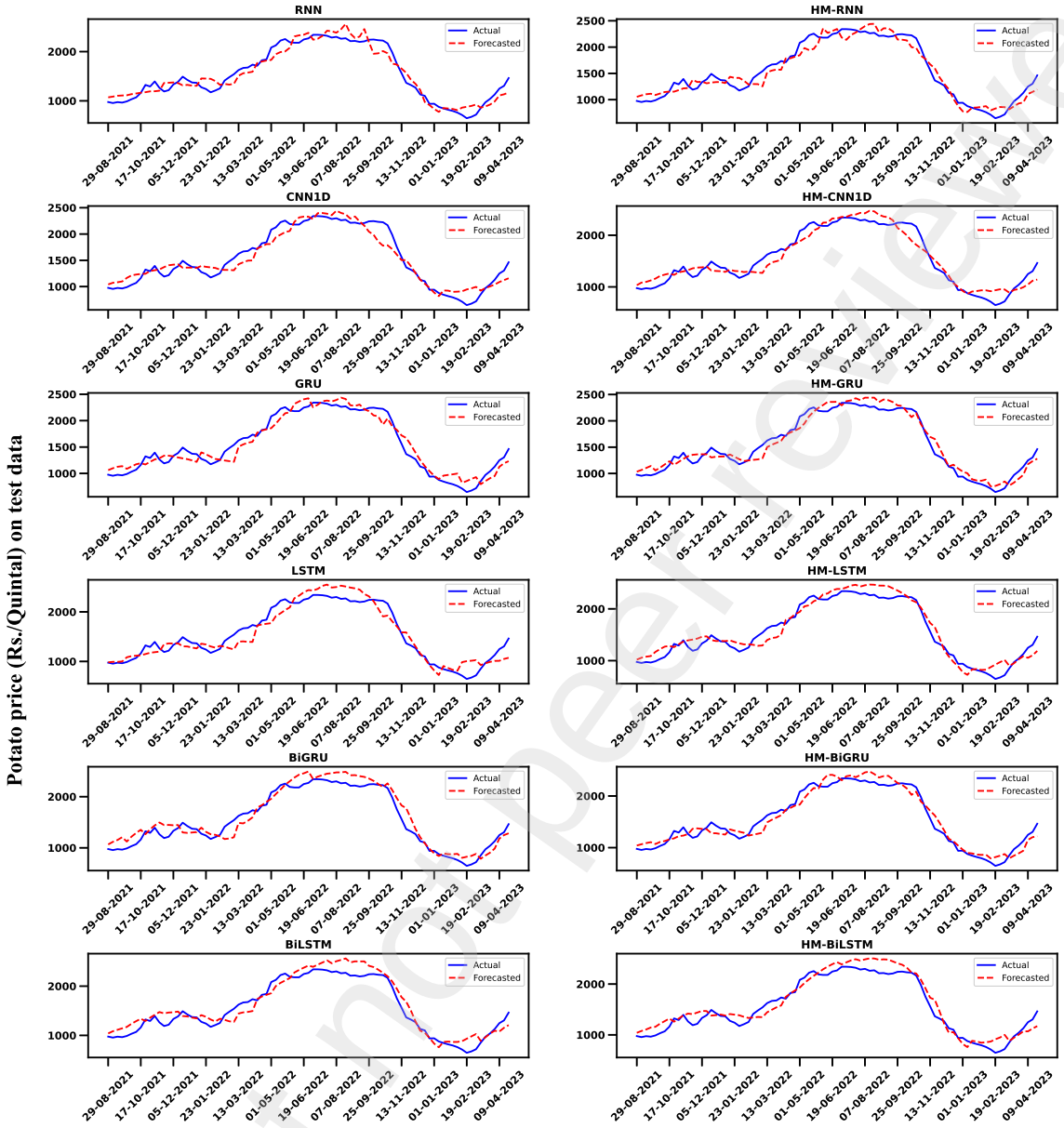


Figure A.12: Four weeks ahead forecasted price series obtained from DL and HM-DL models

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

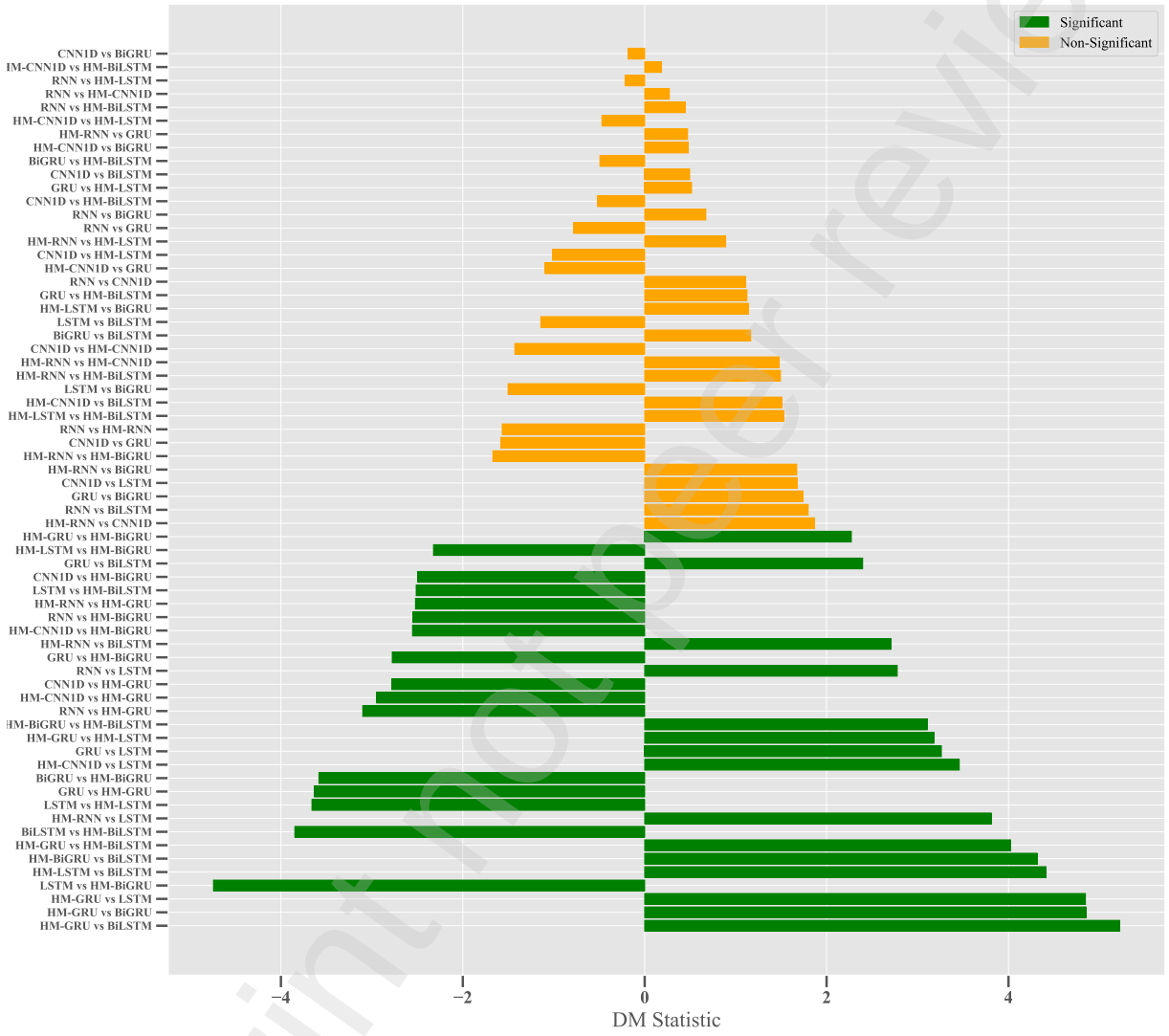


Figure A.13: DM test on potato price series for four weeks ahead forecasts

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

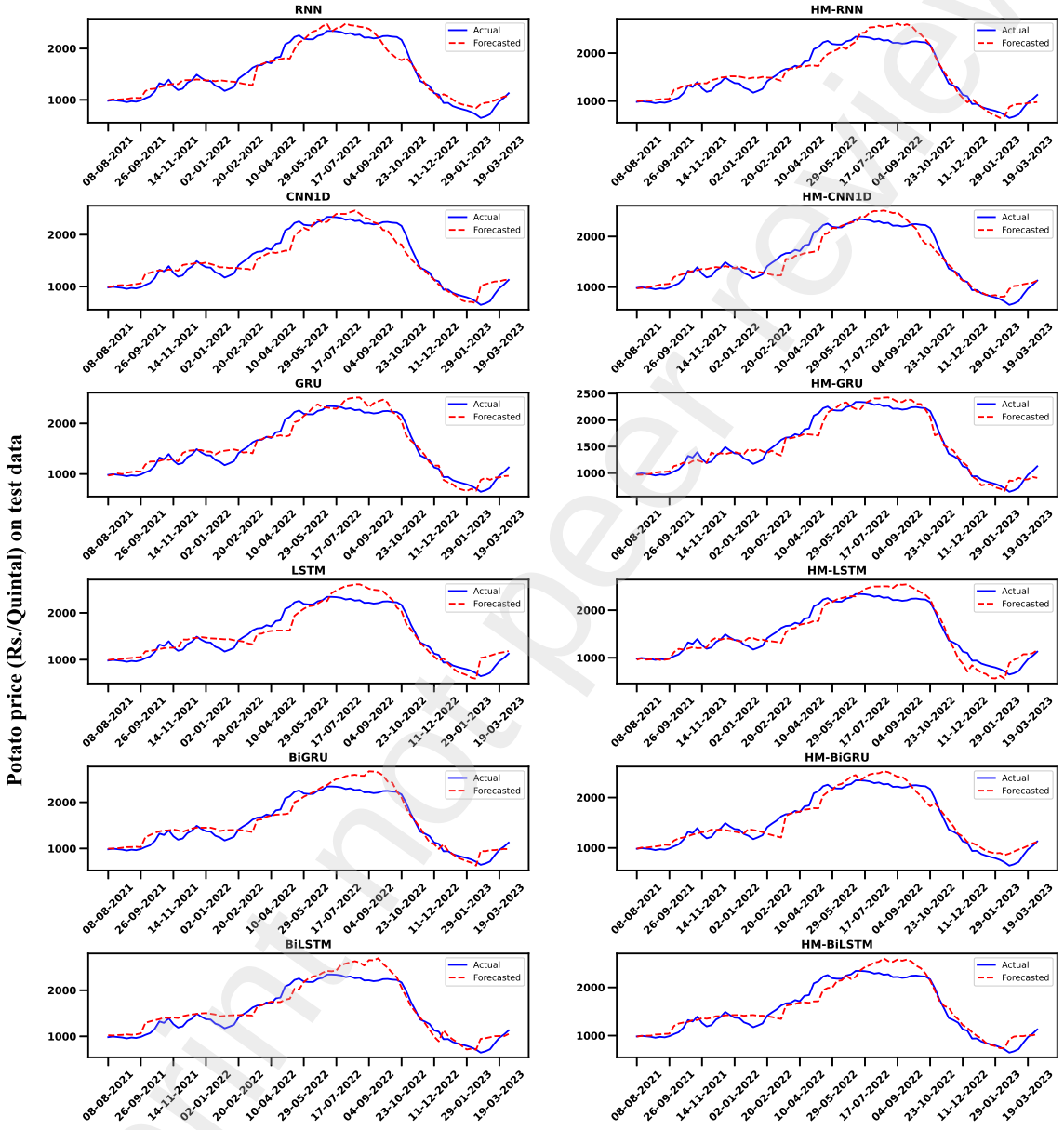


Figure A.14: Eight weeks ahead forecasted price series obtained from DL and HM-DL models

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

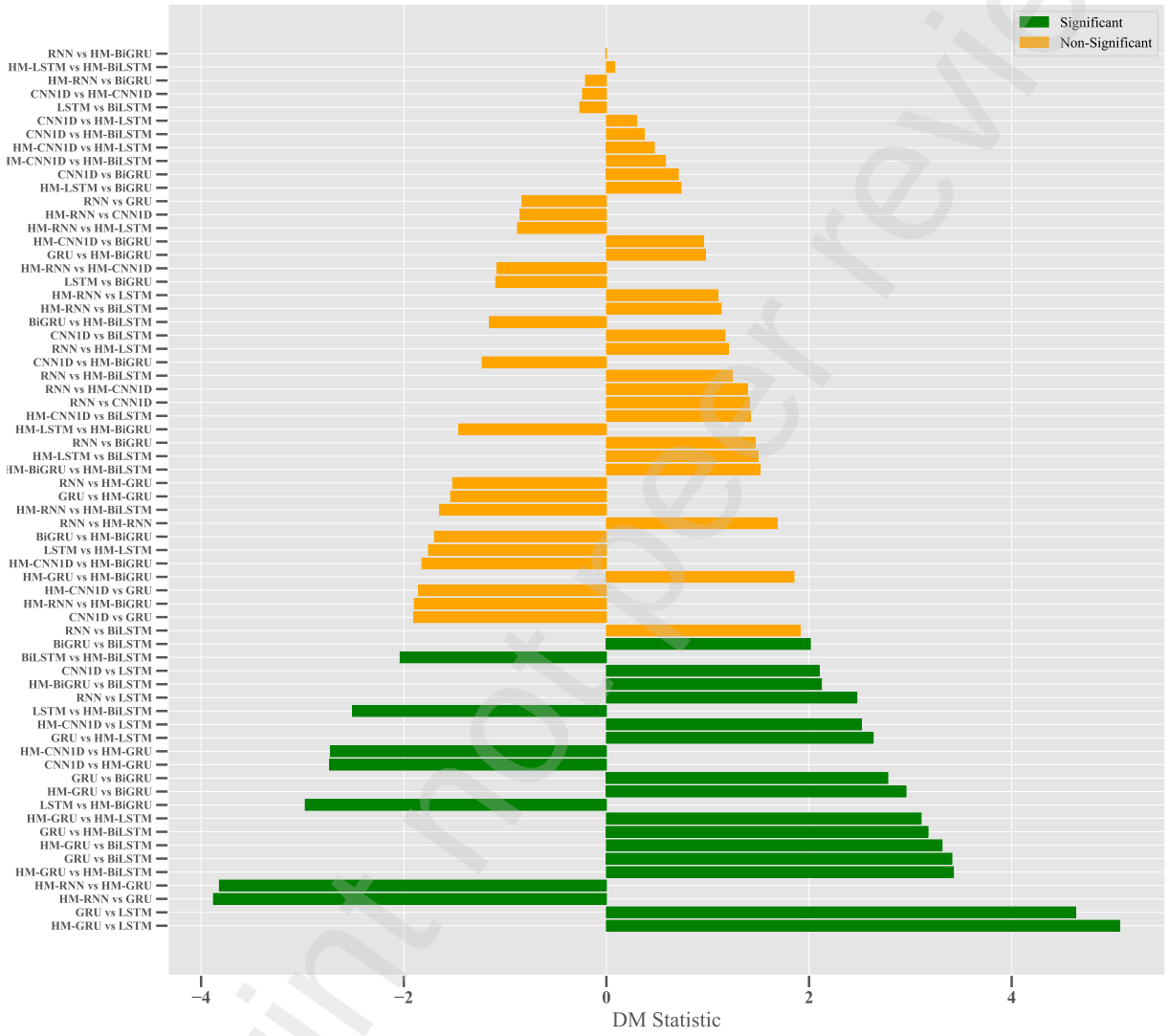


Figure A.15: DM test on potato price series for eight weeks ahead forecasts

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

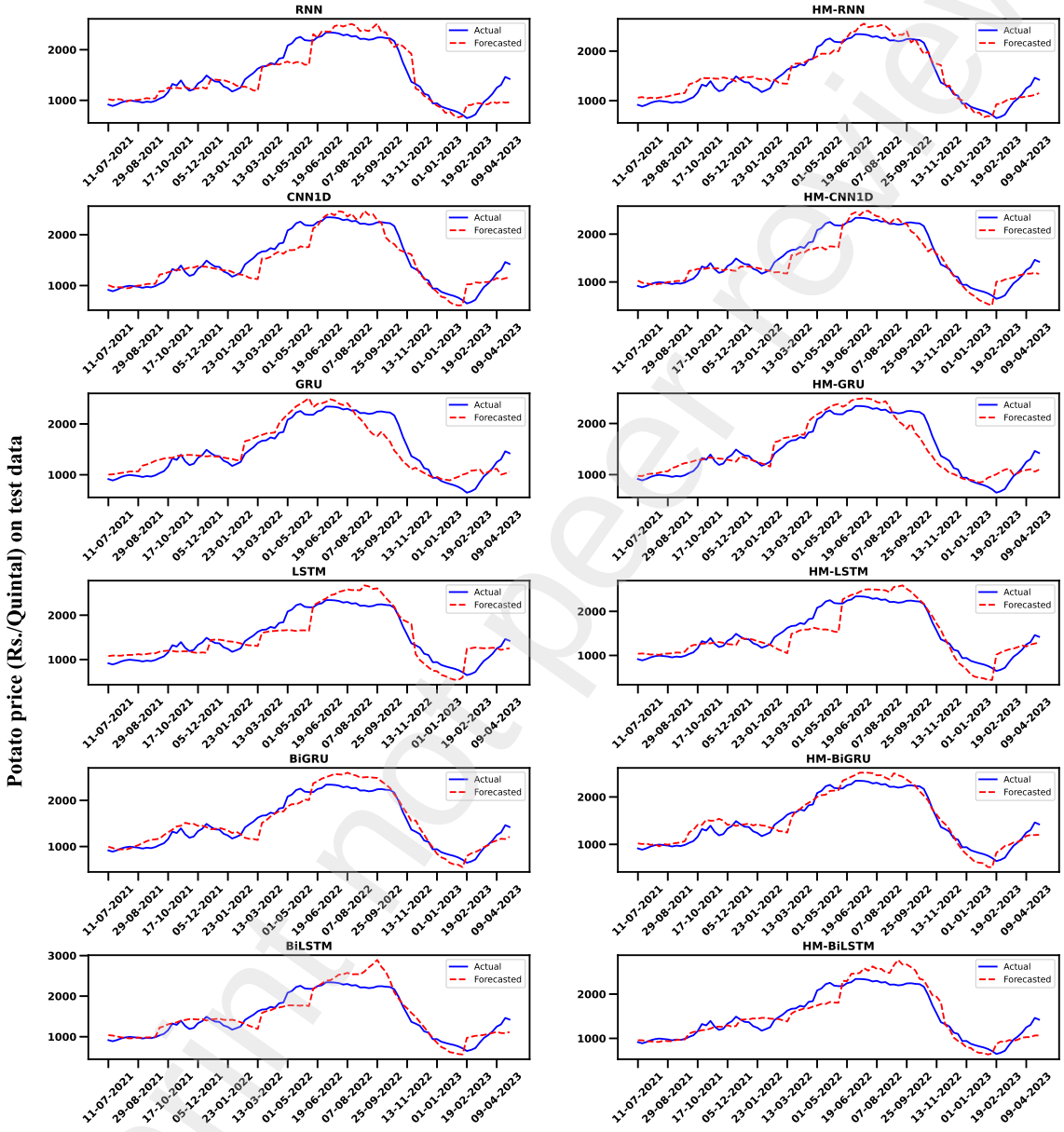


Figure A.16: Twelve weeks ahead forecasted price series obtained from DL and HM-DL models

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

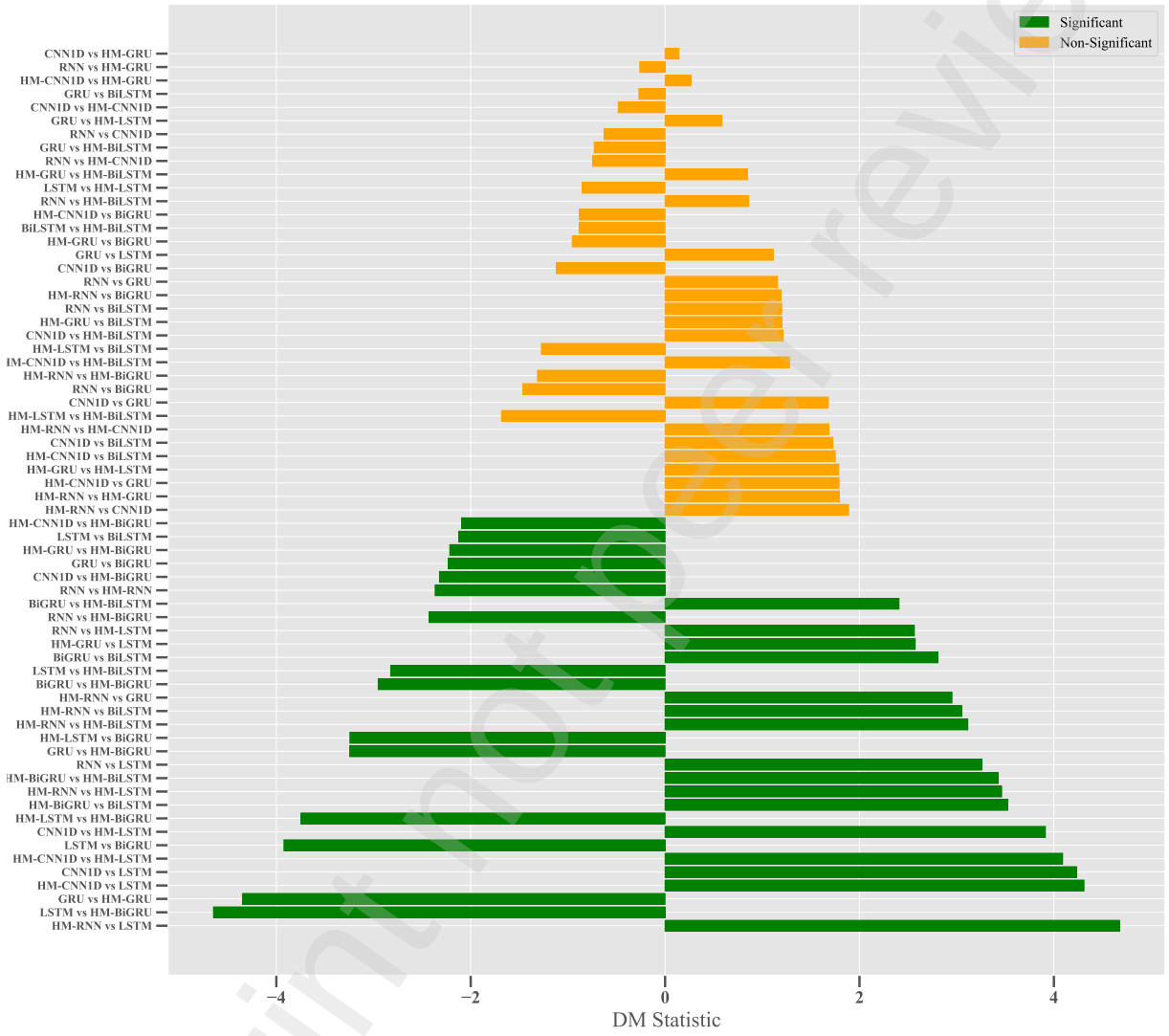


Figure A.17: DM test on potato price series for twelve weeks ahead forecasts

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

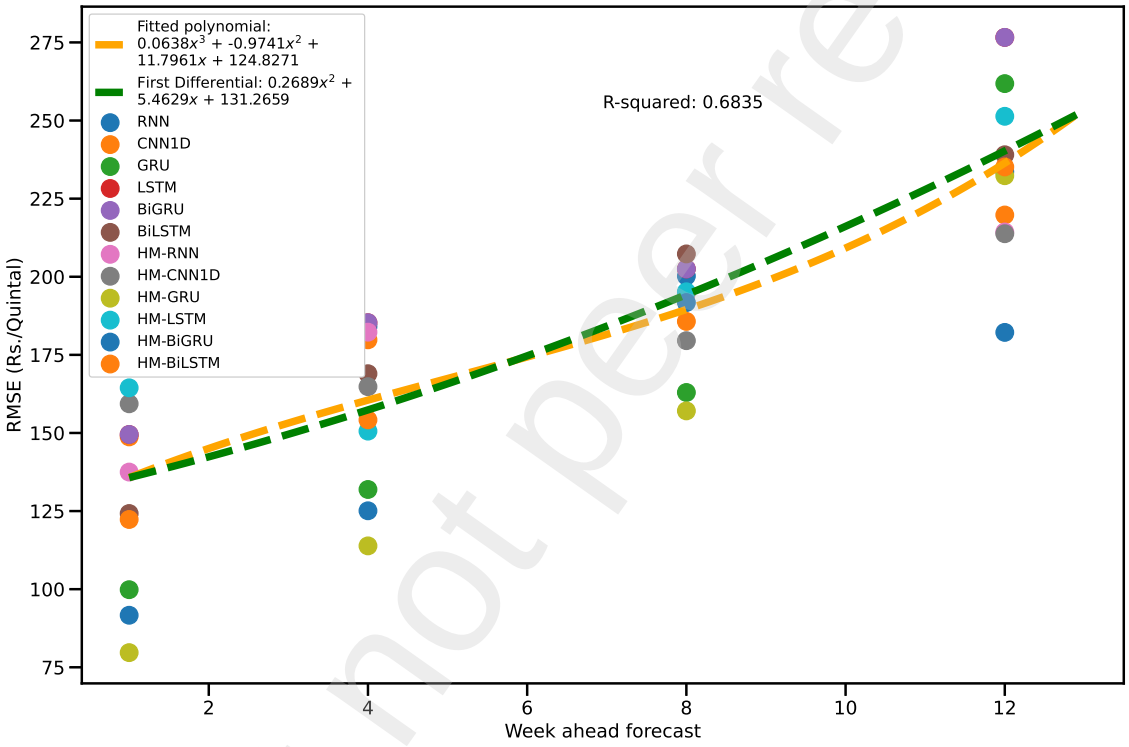


Figure A.18: Relation between RMSE values on weekly ahead forecast

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

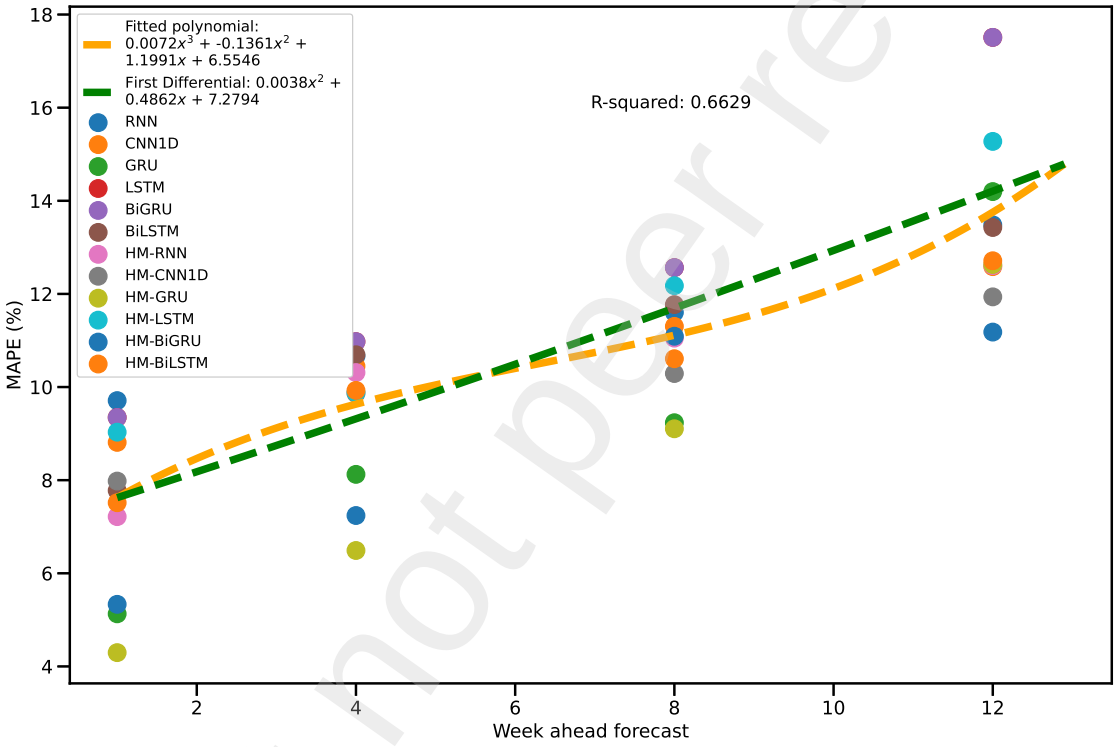


Figure A.19: Relation between MAPE values on weekly ahead forecast

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

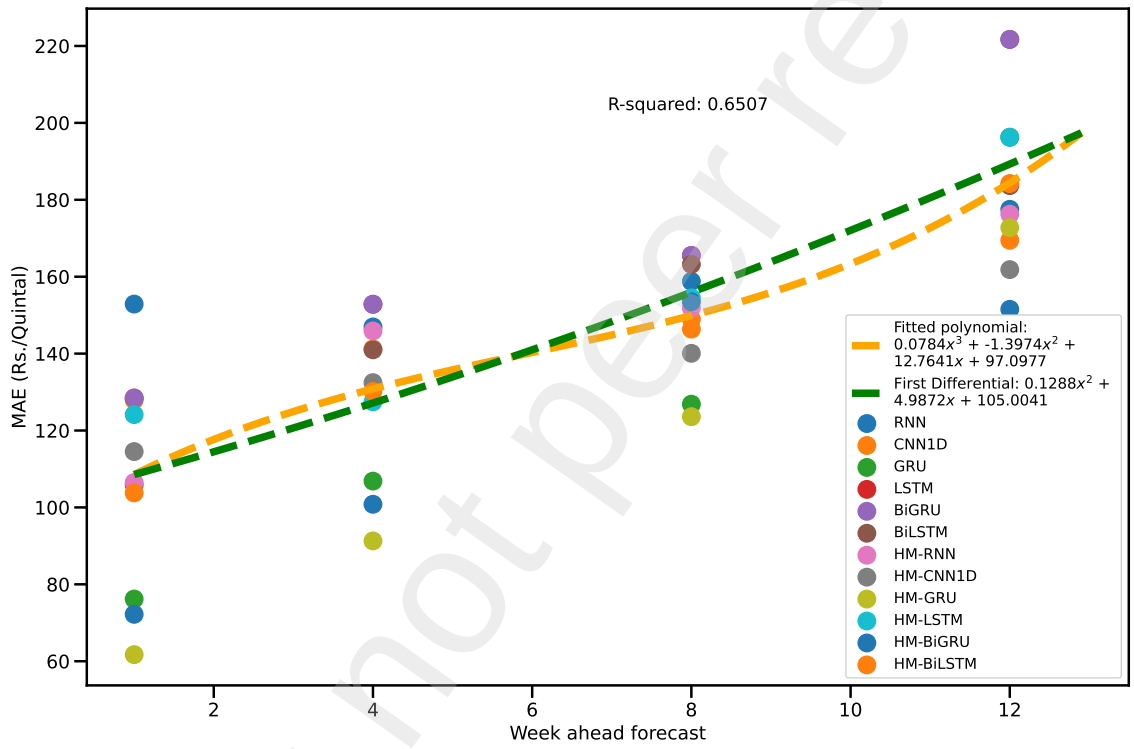


Figure A.20: Relation between MAE values on weekly ahead forecast

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

HP	Ahead forecast	RNN	CNN	LSTM	GRU	BiLSTM	BiGRU	HM-RNN	HM-CNN	HM-LSTM	HM-GRU	HM-BiLSTM	HM-BiGRU
BS	**1-Wk**	128	128	128	128	128	32	128	128	128	128	128	64
	4-Wk	128	128	64	64	64	32	128	128	128	128	64	64
	8-Wk	128	128	32	128	64	128	128	128	32	128	64	64
	12-Wk	64	256	32	128	32	64	128	256	64	128	128	64
HL	**1-Wk**	2	2	1	2	2	2	2	2	2	2	2	2
	4-Wk	2	2	2	2	2	2	2	3	2	2	2	2
	8-Wk	2	2	2	2	2	2	2	3	2	2	2	2
	12-Wk	2	2	3	2	2	2	2	3	2	2	2	2
HU/FS	**1-Wk**	128, 64	128, 64	128	256, 64	256, 64	64, 32	128, 64	64, 32	256, 64	128, 32	256, 64	128, 64
	4-Wk	128, 64	128, 64	128, 32	128, 64	128, 64	64, 32	128, 64	128, 64, 32	512, 128	256, 128	128, 64	128, 64
	8-Wk	128, 64	128, 64	32, 16	128, 64	128, 64	64, 32	128, 64	128, 64, 32	128, 64	256, 128	128, 64	128, 64
	12-Wk	128, 64	256, 64	128, 64, 32	128, 64	128, 64	32, 16	128, 64	256, 128, 64	128, 64	512, 128	256, 128	128, 64

HP:Hyperparameters; HL: No.of hidden layers; HU:No. of hidden units; FS: Filter size; Wk: Week

Table A.5: Optimal hyperparameters for several forecasting models

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4594856

Preprint not peer reviewed